Class 7
Chemistry

Table of Content

1.	Acids, Bases, and Salts	1 - 5
2.	Classification of Elements	6 – 14
3.	Solutions	15 – 24
4.	Chemistry in Daily Life	25 – 36
5.	Air and Oxygen	37 – 42
6.	Fibre to Fabric	43 – 51

1. Acids, Bases and Salts Solutions

LEVEL-1

1. Correct option: (A)
The acid present in stomach for digestion is hydrochloric acid, which is also known as digestive acid.

2. Correct option: (B)
Sodium Hydrocarbonate used for baking bread. It is also known as baking soda.

3. Correct option: (B)
The formula for common salt used daily is sodium chloride (NaCl).

4. Correct option: (A)
Acids characteristically have a sour taste. Examples: lemon (citric acid), vinegar (acetic acid).

5. Correct option: (B)
Most bases have a bitter taste and feel slippery or soapy.

6. Correct option: (C)
Acids are classified into organic acids (from plants and animals) and mineral (inorganic) acids (obtained from minerals).

7. Correct option: (A)
Vinegar contains about 4–8% acetic acid.

8. Correct option: (D)
Ant stings contain formic acid, also called methanoic acid.

9. Correct option: (B)
Citric acid gives citrus fruits their sour taste

10. Correct option: (C)
Lactic acid is produced by bacteria during milk fermentation.

11. Correct option: (D)
Citric acid is the main acid present in lemons.

12. Correct option: (A)
Organic acids occur naturally in living organisms.

Level-II

1. Correct option: (D)
Tartaric acid occurs naturally in tamarind and grapes.

2. Correct option: (B)
Ascorbic acid is Vitamin C.

3. Correct option: (C)
Ascorbic acid is the chemical name of Vitamin C.

4. Correct option: (C)
Spinach contains oxalic acid.

5. Correct option: (C)
Bases have a bitter taste.

6. Correct option: (C)
Bases feel slippery or soapy.

7. Correct option: (B)
Acids and bases neutralize each other.

8. Correct option: (C)
Lime water is a solution of calcium hydroxide

9. Correct option: (D)
Milk of magnesia contains magnesium hydroxide**.**

10. Correct option: (D)
Window cleaners commonly contain ammonium hydroxide.

11. Correct option: (A)
Washing soda is sodium carbonate.

12. Correct option: (A)
Baking soda is sodium hydrogen carbonate ($NaHCO_3$).

13. Correct option: (D)
Calcium hydroxide, sodium carbonate and magnesium hydroxide are commonly considered weak/basic substances at this level

14. Correct option: (A)
Caustic soda is sodium hydroxide (NaOH).
One or more than one

15. Correct option: (A,D)
Soluble bases ,Bases dissolved in water

16. Correct option: (A,D)
Basicity is the number of replaceable H^+ ions**.**

17. Correct option: (A,D)
Acidity of a base is the number of hydroxyl (OH^-) groups.
A. Number of OH^- ions released
D. Number of hydroxyl groups

18. Correct option: (A,B,D)
Metal oxides + water → Bases
Quick lime + water → Slaked lime ,Slaked lime is a base

19. Correct option: (A,B,D)
Direct combination of elements (e.g., $H_2 + Cl_2 \rightarrow HCl$)
Non-metal oxide + water → Acid
$CO_2 + H_2O \rightarrow$ Carbonic acid

20. Correct option: (A) Both statements are true. Oxyacids contain oxygen, and prefixes/suffixes such as hypo-, per-, -ous and -ic depend on the number of oxygen atoms.

21. Correct option: (A)
Mineral acids are called mineral acids because they are prepared from naturally occurring minerals.

22. P→D; Q→ C; R →A; S →B

23. Correct option: (B)
Strong acids completely ionise in water.

24. Correct option: (B)
Most organic acids are weak acids.

25. Correct option: (B)
Since strong acids ionise completely, they produce more hydrogen ions (H^+).

26. A monobasic acid releases one H^+ ion per molecule (e.g., HCl).

27. The five basic tastes are:
Sweet
Sour
Salty
Bitter
Umami (savory)

28. A typical strong base such as NaOH dissociates completely:
$NaOH \rightarrow Na^+ + OH^-$

It produces **two ions**: one positive ion (Na^+) and one hydroxide ion (OH^-).

Properties of acids and bases,salts

1. Correct option: (B)
The indicators tell about the acidic nature of a substance by change the colour when they mix with a substance.

2. Correct option: (B)
An indicator gives different colours in acid and base.

3. Correct option: (C)
The colour of the litmus solution when it is neutral is mauve(pale purple).

4. Correct option: (A)
The colour of the litmus solution when an acid is added is red.

5. Correct option: (B)
The colour of the litmus solution when a base is added is blue.

6. Correct option: (B)
Lichen is the name of the plant from which litmus is extracted.

7. Correct option: (A)
If a drop of the given solution turns blue litmus paper to red, then the given solution will be acid in nature.

8. Correct option: (B)
If a drop of the given solution turns red litmus to blue, then the given solution will be base in nature.

9. Correct option: (A)
In Hindi china indicator is called as Gudhal.

10. Correct option: (B)
Light pink is the colour of the China rose solution.

11. Correct option: (A)
Acid turns China rose solution from light pink to deep pink.

12. Correct option: (C)
Bases turns China rose from light pink to green.

13. Correct option: (C)
Turmeric turns red in basic solution.

14. Correct option: (D)
Turmeric remains yellow in acidic solution.

15. Correct option: (A)
Phenolphthalein is a man made indicator.

16. Correct option: (C)
Phenolphthalein indicator is colourless in acid solution.

17. Correct option: (D)
Phenolphthalein indicator is pink in basic solution.

18. Correct option: (C)
An acid reacts with a base to form a salt and water.

LEVEL-II

1. Correct option: (B)
All metal hydroxides are bases.

2. Correct option: (C)
Those substances which are neither acidic nor basic in nature are called neutral substances.

3. Correct option: (C)
The reaction in which an acid reacts with a base to form salt and water is called neutralization.

4. Correct option: (A)
Heat form of energy evolved in a neutralization reaction.

5. Correct option: (A)
NaCl is the chemical formula of the salt formed in this neutralization reaction of sodium hydroxide and hydrochloric acid.

6. Correct option: (C)
Salt and water are the substances formed by the reaction of an acid with a base.

7. Correct option: (C)
Acid and base is the parent compound of a salt.

8. Correct option: (B)
The first part of the salt's name is derived from the name of the base.

9. Correct option: (A)
The second part of the salt's name is derived from the name of the acid.

10. Correct option: (D)
The salts of hydrochloric acid are called chlorides.

11. Correct option: (C)
The salts of nitric acid are called nitrates.

12. Correct option: (A)
Acidic salts turns blue litmus to red. These salts are formed by the reaction of strong acid and weak base. So the total hydrogen are not replaced by the base in neutralization reaction.

One or more than one option correct type

13. Correct option: (A,B)
A neutralisation reaction is Acid + Base → Salt + Water. H^+ and OH^- ions combine to form water, while the remaining ions form the salt.

14. Correct option: (A,B)
Normal salts are formed when all replaceable H^+ ions of an acid are completely replaced by metal or ammonium ions, i.e., complete neutralisation occurs.

15. Correct option: (A,B and C)
Hydrated salts contain water of crystallisation. Examples are $CuSO_4 \cdot 5H_2O$, $FeSO_4 \cdot 7H_2O$, and $Na_2SO_4 \cdot 10H_2O$. Anhydrous $CuSO_4$ contains no water molecules, so it is not hydrated.

16. Correct option: (A,B and C)
Litmus, methyl orange, phenolphthalein, and red cabbage juice are all **acid-base indicators** that change colour depending on the pH of a solution.

Assertion and reason type:

17. Correct option: (B)
The assertion correctly defines hydrated salts. The reason correctly defines anhydrous salts, but it does not explain why hydrated salts contain water of crystallisation. It merely contrasts hydrated and anhydrous salts.

18. Correct option: (B)
Sodium hydroxide (NaOH) melts at a high temperature (about **318°C**) but does not decompose under ordinary heating. Some other hydroxides such as **$Cu(OH)_2$** and **$Fe(OH)_3$** decompose on heating. Therefore, the reason is true but does not explain why NaOH melts.

19. P → B;Q → A;R → D;S → C

20. Correct option: (B)
Acids react with metals to produce salt and hydrogen gas**.**Example:$Zn + 2HCl \rightarrow ZnCl_2 + H_2 \uparrow$

21. Correct option: (C)
Acids react with metal carbonates to form salt, water, and carbon dioxide gas.
Example:
$\mathbf{CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2 \uparrow}$

22. Correct option: (C)
Acid + Metal → Salt + Hydrogen
Acid + Carbonate → Salt + Water + Carbon dioxide
The common product in both reactions is salt.

23. Answer: 2
Explanation:
Neutralisation involves two ions:
H^+ from the acid
OH^- from the base
They combine to form water:
$H^+ + 0H^- \rightarrow H_20$

24. Answer: 3
Explanation:
Based on the extent of neutralisation, salts are classified into:

1. **Normal salts** – formed by complete neutralisation.
2. **Acidic salts** – formed by partial neutralisation of a polybasic acid.
3. **Basic salts** – formed by partial neutralisation of a polyhydroxy base.

Previous year questions:

1. **Correct option B) II and III only**
This is a **neutralisation reaction** in which an acid reacts with a base to form **salt and water**, releasing heat. **Sodium sulphate** is a **neutral salt**, not an acidic salt.
2. **Correct option:** B) Phenolphthalein – Ascorbic acid – Sulphuric acid – Tartaric acid
Phenolphthalein → Colourless in acids and pink in bases.
3. **Correct option: D) Lime water, Milk of magnesia, Curd**
Phenolphthalein turns pink only in bases, so lime water is suitable.
China rose indicator turns green in bases, so milk of magnesia fits.
Methyl orange turns red in acids, and curd is acidic due to lactic acid.
4. **Correct option:** C) Red – Yellow – Red – No change
Test Tube I – Amla juice
Amla contains Vitamin C (Ascorbic acid).
It is acidic.
Colour: Red
Test Tube II – Detergent solution
Detergent solution is basic.
Colour: Yellow
Test Tube III – Soda water
Soda water contains dissolved carbon dioxide, forming carbonic acid.
It is acidic.
Colour: Red
Test Tube IV – NaCl solution
Sodium chloride solution is neutral.
Colour: No change
5. **Correct option:** All options are correct
A wasp sting consists of formic acid that can be neutralised by rubbing magnesium hydroxide.
True
A wasp sting is acidic due to formic acid.
Magnesium hydroxide (Milk of magnesia) is a base and neutralises the acid.
II. *Ammonium chloride is an acidic salt while sodium acetate is a basic salt.*
True
Ammonium chloride (NH_4Cl) is formed from a strong acid (HCl) and weak base (NH_4OH), so it is acidic.
Sodium acetate (CH_3COONa) is formed from a weak acid (acetic acid) and strong base (NaOH), so it is basic.
III. *The tarnished surface of copper articles can be treated by rubbing lemon on them.*
True
Lemon contains citric acid.
The acid dissolves the green coating of basic copper carbonate, restoring the shine.
IV. *If the soil becomes too acidic, then organic matter is added to the soil which releases bases on decomposition.*
True
Organic matter (compost/manure) decomposes and releases substances that help reduce soil acidity.
Farmers may also add lime to neutralise acidic soil.
6. **Correct option: B)** I, III and IV only
7. **Correct option: A)** Calamine solution and lemon juice
Explanation
Calamine solution is basic, so China rose turns green.
Lemon juice is acidic, so China rose turns magenta.
8. **Correct option: C)** Statement 1 is true but Statement 2 is false.
9. **Correct option: B**) III and IV only
10. **Correct option: B)** It turns red litmus blue and phenolphthalein pink, hence it is a basic solution.

2. Classification of Elements

Solutions

LEVEL-I

1. Correct option: C) Chlorine, Bromine, Iodine

Dobereiner's Law of Triads states that:

- Elements can be grouped in threes (triads) having similar chemical properties.
- The atomic mass of the middle element is approximately the average of the atomic masses of the first and third elements.

2. Correct option: (B)

Antoine Lavoisier, a French chemist, is considered the "Father of Modern Chemistry." his work laid the foundation for the systematic study of elements and their properties. Lavoisier's experiments and studies in the late 18th century helped to establish the concept of chemical elements and the distinction between metals and non-metals based on their properties.

3. Correct option: (D)

Carbon, Nitrogen, and Oxygen can form a triad in the concept proposed by Johann Wolfgang Dobereiner's. They have similar properties, and their atomic masses follow a pattern where the atomic mass of the middle element (Nitrogen) is approximately the average of the atomic masses of the other two elements (Carbon and Oxygen).

4. Correct option: (A)

The law of triads was indeed proposed by Johann Wolfgang Dobereiner, he is a German chemist, in 1829.He noticed that certain groups of three elements had similar chemical properties, and the atomic mass of the middle element was approximately the average of the other two elements in the triad. This observation was the basis of his concept of triads.

5. Correct option: (A)

John Newlands, an English chemist, proposed the law of octaves in 1864. He observed that when elements were arranged in order of increasing atomic masses, their properties repeated every eighth element, similar to the repetition of musical notes in an octave. According to this law, potassium is the eighth element after sodium in the periodic table, and its properties are similar to those of sodium.

6. Correct option: (A)

Elements are arranged in order of increasing atomic masses and observed that certain properties repeated every eighth element. In this case, fluorine is the first element, and the eighth element after fluorine is chlorine. The properties of chlorine are similar to those of fluorine, following the pattern set by Newlands' law of octaves and they also belongs to the same group halogens.

7.Correct option: (C)

The electro positivity of elements defined as the tendency to lose electrons and form positive ions.

Sodium (Na) is a highly electropositive alkali metal with atomic number 11. It readily loses one electron to form a sodium ion (Na^+). Iron and Nickel are transition metals and are less electropositive than sodium. Chlorine is a non-metal and is highly electronegative, that means it tends to gain electrons, making it the least electropositive element among the options.

8.Correct option: (D)

Electro negativity of an element is defined as the ability to attract and hold onto electrons when chemical bond is formed. Chlorine is a halogen, with atomic number 17 near to

Argon (inert gas) and halogens are known for their high electro negativity. They have a strong tendency to attract electrons in chemical bonds, making them highly electronegative. Oxygen is also quite electronegative and comes next after chlorine
Nitrogen and carbon are less electronegative compared to chlorine and oxygen, with carbon generally having lower electro negativity among the two.

9. Correct option: (D)

John Newlands' law of octaves, proposed in 1864, he done early attempt to classify the elements based on their properties. However, it was limited in its applicability as it worked well only up to calcium. Beyond calcium, the pattern of repetition of properties every eighth element breaks down, and the law becomes less accurate in describing the properties of elements.

10. Correct option: (D)

John Newlands proposed the law of octaves in 1864, which was the first attempt to classify the elements based on their properties and atomic masses. He noticed that when elements were arranged in order of increasing atomic masses, certain properties repeated every eighth element, similar to the repetition of musical notes in an octave. While Newlands' law of octaves had limitations, it was a crucial step towards the development of the modern periodic table.

11. Correct option: (A)

John Newlands proposed the law of octaves in 1864; he arranged the elements in order of increasing atomic masses and noticed that certain properties repeated every eighth element. Starting from nitrogen (atomic number 7), the eighth element after it would be phosphorus (atomic number 15).

12. Correct option: (C)

Ruthenium (Ru), Rhodium (Rh), and Palladium (Pd) do not form a Dobereiner triad because their atomic masses do not exhibit the required pattern of the atomic mass of the middle element being the average of the other two. Dobereiner's triad concept was more successful in organizing elements in the early days of chemistry but later it was replaced by Mendeleev periodic table and others.

13. Correct option: (D)

John Newlands arranged the elements in increasing order of their atomic numbers. In his law of octaves, he observed that when elements were arranged in order of increasing atomic masses, certain properties repeated every eighth element, similar to the repetition of musical notes in an octave.

14. Correct option: (C)

A set of three elements in which the atomic mass of the middle element is approximately the average of the atomic masses of the other two elements is known as a "Triad." This concept was proposed by Johann Wolfgang Döbereiner, a German chemist, in the early 19th century.

15. Correct option: (A)

The discovery of the inert gases, also known as noble gases, gave a death blow to Newland's law of octaves. Newland's law of octaves was an early attempt to classify the elements based on their properties and atomic masses. However, the law was limited in its applicability and accuracy.
When the noble gases were discovered in the early 20th centuries (helium, neon, argon, krypton, xenon, and radon), they did not fit into the pattern described by Newland's law of octaves. The noble gases have unique properties, such as being chemically inert and having a full valence electron shell, which makes them distinct from other elements.

16. Correct option: (A)

John Newlands observed that every eighth element has similar properties.

17. Correct option: (B)

Lother Meyer plotted the graph between atomic volume and atomic mass on X and Y axis respectively. This graph is known as the "Meyer's Curve" or "Meyer's Plot" and is an important contribution to the early understanding of periodic trends in the properties of elements.

In the plot, the x-axis represents the atomic volume, which and is typically expressed in units like cubic centimeters per mole ($\frac{cm^3}{mole}$). The y-axis represents the atomic mass of the elements, usually given in atomic mass units (amu).

18. Correct option: (A)

Lothar Meyer's curve, the peaks are occupied by alkali metals. The curve is also known as the "Meyer's Curve" and it illustrates the relationship between atomic volume (molar volume) and atomic mass of elements.

Alkali metals are found in the leftmost column of the periodic table (Group 1). They include elements such as lithium (Li), sodium (Na), potassium (K), and others. These elements have relatively large atomic volumes and high electropositive nature which is why they appear as peaks on the graph.

The graph helps to show the periodicity of elements, and in Meyer's case, it revealed that elements with similar properties tend to occur at regular intervals when they are arranged by increasing atomic mass.

19. C) Halogens *(The ascending (rising) portions of the curve are occupied by halogens.)*

20. Correct option: (B)

The systematic relationship between the order of atomic masses and the repetition of properties of elements is called "Periodicity."It refers to the regular and predictable pattern of chemical and physical properties that occur as you move across a row (period) or down a column (group) of the periodic table.

The periodicity is a result of the elements being arranged in order of increasing atomic number, and this arrangement allows elements with similar electron configurations and valence electron arrangements to be placed together, leading to the repetition of properties at regular intervals.

21. Correct option: (C)

The **descending portions** of Lothar Meyer's curve are occupied by **transition meta**

22. Correct option: (C)

The minimum curve of Lothar Meyer's curve is occupied by the transition metals. The curve, also known as the "Meyer's Curve" represents the relationship between atomic volume (molar volume) and atomic mass of elements.

Transition metals are found in the d-block of the periodic table and include elements such as Iron (Fe), Copper (Cu), Zinc (Zn), Silver (Ag), and Gold (Au), among others. These elements have relatively smaller atomic volumes compared to their atomic masses, which is why they appear at the minimum positions on the graph.

23. Correct option: (A)

Alkali metals are found in Group 1 of the periodic table and include elements such as Lithium (Li), Sodium (Na), Potassium (K), and others. These elements have relatively large atomic volumes compared to their atomic masses, which is why they appear as peaks on the Meyer's curves. The graph helps to demonstrate the periodicity of elements.

24. Correct option: (B)

Ernest Rutherford is known for the nuclear model of the atom, not for developing the periodic table.

LEVEL-II

1. Correct option: (C)

$$\text{Arithmetic mean} = \frac{\left(\textit{Atomic weight of Calcium} + \textit{Atomic weight of Barium}\right)}{2}$$

$$\text{Arithmetic mean} = \frac{\left(40.078 + 137.327\right)}{2} = \frac{177.405}{2} = 88.703$$

The atomic weight of strontium is approximately 87.62.
The pair with an arithmetic mean closest to the atomic weight of strontium is option C) Calcium and Barium with an arithmetic mean of 88.703, which is the closest to 87.62.

2. Correct option: (C)

C) Relation between the properties of similar elements

3. Correct option: (D)

Elements which exhibit both metallic and non-metallic characters are called metalloids. Metalloids are located along the "staircase" in the periodic table, which separates metals from non-metals. They exhibit properties that are intermediate between those of metals and non-metals. Some common examples of metalloids include Silicon, Germanium and Arsenic.

(ii) Answer: Correct option: (D)

The arrangement of elements into groups of three is called the "triad" system of classification. In this system, elements with similar properties were grouped into sets of three based on their atomic weights. The triad system was one of the early attempts at organizing the elements and identifying patterns in their properties.

4. Correct option: (A)

In a Dobereiner triad, the atomic weight of the middle element is approximately equal to the arithmetic mean of the atomic weights of the other two elements.
Given that the atomic weight of 'C' (Carbon) is 40 and the atomic weight of 'E' is 137, we can find the atomic weight of 'H' (Hydrogen) using the arithmetic mean:

$$\text{Atomic weight of 'H'} = \frac{\left(\textit{Atomic weight of 'C'} + \textit{Atomic weight of 'E'}\right)}{2}$$

$$\text{Atomic weight of 'H'} = \frac{\left(40 + 137\right)}{2}$$

$$\text{Atomic weight of 'H'} = \frac{177}{2}$$

Atomic weight of 'H' = 88.5
Therefore, the atomic weight of 'H' is 88.5.

5. Correct option: (D)

A) The law of octaves was the first logical attempt to classify elements on the basis of atomic weights. Newland observed that every eighth element in the list had properties similar to the first, which he called the "law of octaves."
B) Periodicity of elements was recognized for the first time. Newland's work led to the recognition of periodicity in the properties of elements, indicating that elements with similar properties recurred at regular intervals.
So, both statements A and B are correct.

6. Correct option: (D)

According to Dobereiner's law of triads, chemically similar elements are arranged in the order of their atomic weights in groups of three elements.

Dobereiner observed that certain groups of three elements with similar chemical properties existed, and the atomic weight of the middle element was approximately the average of the other two elements in the triad. This implied that elements with similar chemical behaviors were found to have a relationship in terms of their atomic weights.
modern periodic table, with potassium (K) having a higher atomic weight than argon (Ar).

7. (D) Chemically similar, atomic weights.

Explanation

- Johann Wolfgang Döbereiner's law of triads states that when **chemically similar** elements are arranged in increasing order of their **atomic weights** (atomic masses), they form groups of three called triads. In each triad, the atomic weight of the middle element is approximately equal to the arithmetic mean (average) of the atomic weights of the other two elements.

One or more than one option

8. Correct option: (A)

According to Newland's law of octaves, the elements having similar properties are arranged in groups of eight, where the eighth element shows properties similar to the first. The correct option from the given choices is:
A) Lithium, Sodium, Potassium

9. Correct options: (A, B, D)

The correct statement(s) are:
A) Law of octaves could be best applied only up to the element Calcium.
B) Newly discovered elements could not fit into the octave structure.
D) Law of octaves was based on atomic weights.
A) Law of octaves could be best applied only up to the element Calcium: This statement is correct.
Newland's law of octaves was based on arranging elements in order of increasing atomic weights. He observed that elements with similar properties recurred every eighth element, similar to the notes in an octave of music. However, the pattern broke down after calcium, and the law was not applicable beyond that point.
B) Newly discovered elements could not fit into the octave structure: This statement is correct.
Newland's law of octaves was based on the existing elements known during his time. As new elements were discovered beyond calcium, they did not fit into the repeating pattern of the octaves, which was one of the limitations of his classification.
D) Law of octaves was based on atomic weights: This statement is correct.
Newland's law of octaves was primarily based on arranging elements in the order of their increasing atomic weights. He noticed that elements with similar properties occurred at regular intervals, much like the repeating pattern of musical octaves.

10. Correct option: (B, C and D)

Group VIII consists of 9 transition elements and they are further arranged into three triads based on similarities in their physical and chemical properties.
These triads are
- Iron, Cobalt, and Nickel
- Ruthenium, Rhodium, and Palladium
- Osmium. Iridium and Platinum

11. Correct options (A, C, D)

Lother Mayer curves represent the periodic variation of properties of elements with increasing atomic numbers. The peaks in these curves represent the elements with maximum values of a particular property. To identify the elements that are not at the peak of the Lother Mayer curves, we need to check the periodic properties of the given elements:

A) V (Vanadium): Vanadium is not at the peak of Lother Mayer curves. It is located in the middle of the d-block transition metals in the periodic table.
C) Al (Aluminum): Aluminum is not at the peak of Lother Mayer curves. It is a p-block element located in Group 13 of the periodic table.
D) Zn (Zinc): Zinc is not at the peak of Lother Mayer curves. It belongs to period 4.

12. Correct options: (A, C)

The correct statement is:
A) Sodium is a strongly electropositive metal.
C) Chlorine is a strongly electronegative non-metal.
Explanation:
A) Sodium is a strongly electropositive metal: This statement is correct. Sodium (Na) is an alkali metal located in Group I of the periodic table. Metals in Group I are highly electropositive, meaning they readily lose electrons to form positively charged ions (cations).
C) Chlorine is a strongly electronegative non-metal: This statement is correct. Chlorine is a non-metal and is one of the most electronegative elements on the periodic table. It readily gains electrons to form negatively charged ions (anions) in chemical reactions.
So, the correct statements are Sodium is a strongly electropositive metal, and Chlorine is a strongly electronegative non-metal.

13. Correct option: (B)

The classification proposed by Dobereiner, known as Dobereiner's law of triads, was based on the atomic masses of elements. He observed that certain groups of three elements had similar chemical properties, and the atomic weight of the middle element was approximately the average of the other two elements in the triad. This implied that elements with similar chemical behaviors were found to have a relationship in terms of their atomic masses. However, this classification was later replaced by more comprehensive periodic table arrangements, such as Mendeleev's periodic table, which were based on atomic weights but also considered other properties and trends.

ASSERTION AND REASON TYPE

14. Dobereiner's triad classification...

- Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- Explanation: Dobereiner could identify only a few triads from the known elements at that time because many known elements could not fit into the triad system, and forcing certain elements into triads grouped chemically dissimilar elements together. Thus, the classification was not successful.

15. Classification of elements helps in better understanding...

- Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- Explanation: Classifying elements simplifies their study by grouping similar elements together. This grouping works because it relates their physical and chemical behavior to underlying common properties (like atomic mass or atomic number).

Matrix Matching Type

16. a) Lavoisier classified → R) Metals and Non-metals (Lavoisier grouped known elements into metals and non-metals.)
b) Dobereiner classified → S) Triads (Dobereiner arranged elements in groups of three called triads.)

c) Lothermeyer → P) Graphs between atomic volume Vs atomic mass (Lothar Meyer plotted atomic volume against atomic mass to show periodicity.)
d) Mean atomic mass of Ca, Sr, Ba →. T) 88.1 for Sr/average calculation).

17. Newlands' Law of Octaves (Assuming starting from element 1):

a) The eighth element from 'F' (Fluorine) → Q) Cl (Fluorine and Chlorine have similar properties.)
b) The eighth element from 'O' (Oxygen) → R) S (Oxygen and Sulfur share a column.)
c) The eighth element from 'B' (Boron) → S) Al (Boron and Aluminum are in the same group.)
d) The eighth element from 'N' (Nitrogen) → P) P (Nitrogen and Phosphorus share a group.)

Comprehension Type

18. Answer: B) Metals
Explanation: These physical properties (malleability, ductility, high conductivity) uniquely define metallic elements.

19. Answer: A) Non-metals
Explanation: Non-metals break easily when struck (brittle) and lack free electrons, making them poor conductors.

20. Answer: C) Metalloids
Explanation: Metalloids (or semimetals) display intermediate physical and chemical properties between metals and non-metals.

Integer Type

21. Answer: 3
- Explanation: By definition, a "triad" refers to a set or group of 3 elements.

22. Answer: 8th (or 8)
- Explanation: Newlands' Law of Octaves states that when elements are arranged in increasing order of atomic mass, every 8th element shows properties similar to the first one (similar to musical octaves).

Mendeleev's periodic table

Level-I

1. B) Mendeleev
Explanation: Dmitri Mendeleev provided the most significant contribution by formulating the Periodic Law and creating a comprehensive periodic table based on fundamental elemental properties and chemical periodicity.

2. C) Atomic weight
Explanation: Mendeleev's Periodic Law states that the properties of elements are a periodic function of their atomic weights (masses).

3. C) 63
Explanation: Exactly 63 elements were discovered and known at the time Mendeleev constructed his periodic table in 1869.

4. D) Both 'B' & 'C'
Explanation: An anomalous pair consists of elements where an element with a higher atomic weight precedes an element with a lower atomic weight. Both Cobalt (58.9) before Nickel (58.7) and Argon (39.9) before Potassium (39.1) violate the increasing atomic weight sequence.

5. B) Groups
Explanation: In the periodic table, vertical columns are defined as groups.

6. B) Periods
Explanation: Horizontal rows in the periodic table are defined as periods.

7. D) Scandium
Explanation: Mendeleev predicted "Eka-Boron," which was later discovered and named Scandium (Sc).

8. A) Gallium
Explanation: Mendeleev predicted "Eka-Aluminium," which was later discovered and named Gallium (Ga).

9. C) Germanium
Explanation: Mendeleev predicted "Eka-Silicon," which was later discovered and named Germanium (Ge).

10. C) 7
Explanation: Mendeleev's table had **7 periods**.

11. C) To accommodate undiscovered elements
Explanation: Mendeleev intentionally left blank spaces for elements that had not yet been discovered so that elements with similar chemical properties remained aligned in the same group.

12. B) Roman numerals
Explanation: Groups in Mendeleev's periodic table were denoted by Roman numerals (I to VIII).

13. B) 8
Explanation: The original Mendeleev periodic table consisted of 8 groups (I to VIII).
Note: Group 0 was added later for noble gases.

14. C) Arabic numerals (1 to 7)
Explanation: Periods in Mendeleev's table were designated using Arabic numerals (1 to 7).

15. C) 7
Explanation: Mendeleev's periodic table contains 7 periods (horizontal rows).

16. C) One
Explanation: The Sanskrit prefix *Eka* means "one". It indicated that the unknown element was located one place below the reference element in the same group.

Level-II

Single Option Correct Type

1. D) Correction atomic weight of some elements
Explanation: Correcting erroneous atomic weights (such as Beryllium, Gold, Platinum) was a major achievement/advantage of Mendeleev's table, not a defect or limitation.

2. A) Argon – Potassium
Explanation: Argon (Ar, atomic weight 39.9) was placed before Potassium (K, atomic weight 39.1), creating an inversion of atomic weight.

3. B) It has no fixed position
Explanation: Hydrogen exhibits properties resembling both alkali metals (Group I) and halogens (Group VII), making its exact placement ambiguous.

4. C) Tellurium has higher atomic weight but is placed before iodine
Explanation: Tellurium (atomic mass 127.6) was placed before Iodine (atomic mass 126.9) to group elements with similar chemical properties together.

5. B) Their position could not be explained
Explanation: Isotopes have different atomic weights but identical chemical properties. If arranged purely by atomic mass, isotopes of the same element would require separate positions, which Mendeleev's table could not accommodate.

6. C) Sub-groups A and B
Explanation: Mendeleev divided groups into sub-groups A and B. As a result, chemically distinct elements (like alkali metals in IA and coinage metals in IB) were grouped together.

7. C) Li and Cu
Explanation: Lithium (Li) is an extremely reactive alkali metal (Group IA), while Copper (Cu) is a transition metal (Group IB). Placing them in the same main group is an example of grouping dissimilar elements.

8. C) Chlorine is a non-metal, manganese is a metal
Explanation: Chlorine (a halogen non-metal in VIIA) and Manganese (a transition metal in VIIB) have drastically different physical and chemical characteristics despite sharing Group VII.

One or More Than One Option Correct Type

9. A, B, D (Li, Na, K)
Explanation: Lithium (Li), Sodium (Na), and Potassium (K) are Group 1 alkali metals. Calcium (Ca) is an alkaline earth metal (Group 2).

10. A, B, C (Fe, Co, Ni; Ru, Rh, Pd; Os, Ir, Pt)
Explanation: Group VIII of Mendeleev's table contains nine transition elements arranged in three horizontal groups of three elements known as transitional triads.

11. A, C, D (Argon – Potassium; Tellurium – Iodine; Cobalt – Nickel)
Explanation: All three pairs represent inverted atomic mass order: Ar(39.9) > K(39.1), Te(127.6) > I(126.9), and Co(58.9) > Ni(58.7).

Assertion and Reason Type

12.Answer: C) Assertion is true, but Reason is false.

Mendeleev grouped elements with **similar properties**, making the study of elements easier. Therefore:

Assertion → **True**, Reason → **False**

13. (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

- Explanation: Mendeleev corrected atomic weights using the formula Atomic Weight = Equivalent Weight × Valency. For example, Beryllium's valency was corrected from 3 to 2, changing its calculated atomic mass from 13.5 to 9.

Matrix Matching Type

14. **a → R, b → P, c → S, d → Q**
a) Hydrogen → R) 1st period (Period 1) **b**) Lithium → P) 2nd period (Period 2)
c) Sodium → S) 3rd period (Period 3) **d)** Potassium → Q) 4th period (Period 4)

Comprehension Type

15. B) Atomic weight
Mendeleev arranged elements primarily in increasing order of their atomic weights.

16. C) 63
Explanation: There were 63 elements discovered and known at the time.

17. C) Atomic weights
Explanation: Mendeleev's Periodic Law states that properties are periodic functions of their atomic weights.

Integer Type

18. Explanation: The modified (or extended) Mendeleev periodic table contains 9 groups (Group 0 to Group VIII).

19. Explanation: The modified Mendeleev periodic table contains 7 periods.

3. Solutions

LEVEL-I

1. Correct option: (C)

A homogeneous mixture of two or more components is called a solution. In a solution, the compounds are evenly distributed at the molecular or atomic level, resulting in a uniform appearance and properties throughout the mixture.

2. Correct option: (A)

If a solution consists of two components, it is called a binary solution. In a binary solution, there are two different substances dissolved in each other, creating a homogeneous mixture with a uniform composition.

3. Correct option: (A)

The substance which is relatively less in quantity in a solution is called the solute. The solute is present in a smaller amount compared to the solvent in the solution.

4. Correct option: (C)

If a solution is composed of three components, it is called a ternary solution. The term "ternary" indicates the presence of three different components dissolved in each other to form a homogeneous mixture with a uniform composition.

5. Correct option: (B)

There is only one phase present in a true solution. A true solution is a homogeneous mixture where the solute is uniformly dispersed in the solvent. Since the solute particles are distributed evenly at the molecular level, there is no visible distinction between the solute and solvent, resulting in a single-phase system.

6. Correct option: (D)

All solutions are characterized by all of the following:

A) **Absence of settling**: Solutions are homogeneous mixtures in which the solute particles are evenly distributed throughout the solvent, and there is no settling or separation of components over time.

B) **Molecular or ionic state of subdivision of the components**: In a solution, the solute is present in molecular or ionic form, that is the particles are individually dispersed.

C) **Homogeneity**: Solutions are uniform in composition throughout, meaning they have the same properties and composition at every point within the mixture. There are no visible differences or phase boundaries between the components.

7. Correct option: (B)

Solution of water and sugar is commonly known as syrup.

8. Correct option: (C)

The solute present in seltzer solution is carbon dioxide (CO_2). Seltzer is a type of carbonated water, and the carbon dioxide gas dissolved in the water that gives it its fizziness.

9. Correct option: (A)

Solution of water and salt is called brine. Brine is commonly used for pickling and preserving food, as well as in various industrial processes.

10. Correct option: (B)

An alloy is an example of a solid-solid type solution. Alloys are homogeneous mixtures of two or more elements at least one of the components is a metal. The atoms of the different elements mix together forming a solid solution. Common

examples of alloys include brass (copper and zinc), stainless steel (iron and chromium), and bronze (copper and tin).

11. Correct option: (C)

Soda water is an example of a gas in liquid type of solution. It is a mixture of carbon dioxide (gas) dissolved in water (liquid). The carbon dioxide gives soda water its characteristic fizziness and is present as a gas dissolved in the liquid.

12. Correct option: (A)

If air is taken as a binary solution, the solvent is considered to be nitrogen (N_2). Air is primarily composed of nitrogen (approximately 78% by volume) and oxygen (approximately 21% by volume), along with trace amounts of other gases such as carbon dioxide, argon, and others. In this context, nitrogen (N_2) is considered the solvent, and oxygen (O_2) along with the other trace gases are considered the solutes in the binary solution.

13. Correct option: (C)

Oil in water

A heterogeneous mixture is a mixture in which the components are not uniformly distributed and can be distinguished as separate phases within the mixture. Among the options given, "Oil in water" is a heterogeneous mixture. When oil is mixed with water, the two substances do not fully dissolve into each other, and oil droplets remain separate from the water, creating visible distinct separate phases.

14. Correct option: (C)

If the quantity of solute is more in a solution, then the solution is said to be a concentrated solution. In a concentrated solution, a relatively large amount of solute is dissolved in a given quantity of solvent. The concentration of the solute is higher compared to other types of solutions.

15. Correct option: (B)

A solution in which alcohol is used as the solvent is known as an "Alcoholic" solution. In this the term "alcoholic" refers to a solution where alcohol serves as the medium for dissolving one or more solutes. These solutions are common in various industries, for example, alcoholic beverages like wine, beer, and spirits are examples of alcoholic solutions where alcohol is the solvent.

LEVEL-II

1. Correct option: (C)

The correct statement is C) Solutions are homogeneous mixtures.

A solution is defined as a type of mixture in which the solute is uniformly distributed and dissolved in another substance. The particles of the solute mix completely with the particles of the solvent, creating a uniform composition throughout the mixture. Because of this uniform distribution, solutions are considered homogeneous mixture.

2. Correct option: (B)

Camphor in air is an example of a solid (camphor) dispersed in a gas (air). When you place a piece of camphor in the open air, it gradually evaporates and forms a gaseous phase around it, which you can observe as peculiar camphor smell. The solid camphor particles disperse into the air, making it an example of a solid in a gas mixture.

3. Correct option: (C)

Dissolution of glucose in water is correct answer among the given options.

A physical change is defined as a change in physical properties of a substance without disturbing its chemical properties. When glucose is added to water, it gets uniformly distributed in the water, forming a homogeneous mixture. However, the chemical composition of glucose remains the same; it does not undergo any chemical reaction during the dissolution process.

On the other hand, other options involve burning or combustion processes, which are chemical changes. During these processes, the substances undergo chemical reactions and are converted into different compounds.

4. Correct option: (A)

Gas in solid is correct option among all.

Occlusion of hydrogen on palladium is an example of gas (hydrogen) being trapped within the solid structure of palladium. This process is known as "occlusion," and it represents a gas (hydrogen) being incorporated into the lattice structure of the solid (palladium) without forming a chemical compound.

5. Correct option: (C)

Ethanol dissolved in water is the correct option among all.

A liquid-in-liquid type solution, also known as a homogeneous mixture, is a solution where one liquid is uniformly dispersed and dissolved in another liquid. Among the options provided, ethanol dissolved in water is a liquid-in-liquid type solution. When ethanol (a liquid) is mixed with water (liquid), they form a homogeneous mixture with no visible boundaries between the two components.

6. Correct option: (B)

The mixture of benzene and toluene is an example of a liquid-in-liquid type solution. When benzene and toluene are mixed together, they form a homogeneous mixture, with one liquid (benzene) dissolved and uniformly dispersed in the other liquid (toluene).

7. Correct option: (C)

A homogeneous mixture is defined as a mixture in which the components are uniformly distributed and cannot be visibly distinguished. Among the options provided, "Salt in water" is homogeneous in nature. When salt is dissolved in water, it completely dissolves and forms a homogeneous solution, and you cannot see separate salt particles within the water. The salt molecules disperse evenly throughout the water, resulting in a uniform mixture.

8. Correct option: (C)

A heterogeneous mixture is defined as a mixture in which the components are not uniformly distributed, and the different substances can be visibly distinguished. Among the options provided, "Oil in water" is a heterogeneous mixture. When oil is added to water, it does not dissolve but forms separate droplets that can be seen floating on the surface or dispersed throughout the water.

9. **Answer: B) Solutions are pure substances** AND **D) Preparation of a solution is a chemical change**

- *Explanation:* * Solutions are *mixtures*, not pure substances, because they consist of two or more distinct chemical entities (solute and solvent) that retain their individual properties.
  - Preparing a standard solution (like dissolving salt or sugar in water) is a *physical change*, not a chemical change, because no new chemical bonds are created and the components can be separated by physical means like evaporation.

- Statements A and C are true facts about solutions.

10. Answer: B, and C are correctly matched.

- **B) Mixture of oxygen and nitrogen:** Air is a classic example of a **Gas in Gas** solution — Correct.
- **C) Sugar dissolved in water:** Sugar is a solid solute in a liquid water solvent (**Solid in Liquid/Water** — Correct).

Assertion and Reason Type

11. **Assertion (A):** A dilute solution contains a relatively small amount of solute compared to the solvent.

Reason (R): In a dilute solution, the solvent is present in a much larger proportion than the solute.

- **Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**
- *Explanation:* By definition, a dilute solution has a very low concentration of solute because the quantity of the solvent drastically overpowers it. The reason perfectly explains why the solution is considered dilute.

12. **Assertion (A):** A homogeneous mixture has components that are not uniformly distributed throughout the mixture.

Reason (R): Sand and water is an example of a heterogeneous mixture.

- **Answer: (D) Assertion is false, but Reason is true.**
- *Explanation:* The Assertion is false because a *homogeneous* mixture has components that **are** uniformly distributed throughout (like salt water). The Reason is true because sand does not dissolve in water and settles out, making it a classic example of a heterogeneous mixture.

13. Correct Matching is: a-R, b-S, c-Q, d-P

a) Correct option is R.

A binary solution refers to a type of solution that contains two components: a solute and a solvent. The solute is the substance that gets dissolved in the solvent to form the solution. The solvent is the component in which the solute dissolves resulting a homogenous mixture.

b) Correct option is S.

The word “Ternary" indicates that there are three components present, and it doesn't specify the nature of these components or the exact type of mixture.

Ternary Solution: This is a homogeneous mixture where three substances are dissolved in a solvent. For example, a solution containing water as solvent and dissolved salts (sodium chloride and potassium nitrate) could be considered a ternary solution.

c) Correct option is Q.

It is defined as a homogeneous mixture where four substances are dissolved in a solvent. For example, a solution containing water as solvent and dissolved salts (sodium chloride, potassium nitrate, and calcium sulfate) could be considered a quaternary solution.

d) Correct option is P.

In a dilute solution, the amount of solute (the substance being dissolved) is relatively small compared to the amount of solvent (the substance doing the dissolving). Dilution occurs when additional solvent is added to a solution, reducing the concentration of the solute. This can be achieved by adding more solvent to an existing solution or by mixing a concentrated solution with more solvent.

14. Correct Matching is: a-Q, b-S, c-P, d-R

a) Correct option is Q.

A solution of water and sugar is commonly named as a "sugar solution" or a "sugar-water solution." In this solution, water is the solvent (the substance that does the dissolving), and sugar is the solute (the substance being dissolved).

b) Correct option is S.

A water and salt solution is commonly called a "Brine" or "salt solution."

When salt (example NaCl) dissolves in water, it forms a homogeneous mixture where the water acts as the solvent and the salt as the solute. The resulting solution is formed.

c) Correct option is P.

It is the solution made by water and carbon dioxide, resulting in a bubbly and effervescent beverage. The carbonation process involves dissolving carbon dioxide gas into the water, creating carbonic acid (H_2CO_3).

d) Correct option is R.

When ammonia gas is dissolved in water, the resulting solution formed is called "ammonia solution" or "aqueous ammonia." This solution is commonly used in various applications, such as cleaning, fertilizers, and as a source of ammonia in chemical reactions. $NH_4OH(aq) \rightleftharpoons NH_4(aq) + OH_(aq)$

In brine solution, the solute is NaCl (sodium chloride). Brine is a solution of salt (NaCl) in water.

15. Correct option: (A)

The solute (NaCl) dissolves in the solvent (water) to form a homogeneous mixture known as brine.

16. Correct option: (D)

Gas, gas - This is the correct option since air is a mixture of different gases, with no solid or liquid components involved.

17. Correct option: (C)

A solution is a homogeneous mixture of a solute dissolved in a solvent. Let's examine each option:

A) **Milk** - Milk is not a solution. It is an emulsion, which is a type of colloid. It consists of small droplets of fat and protein dispersed in water.

B) **Cheese** - Cheese is not a solution either. It is a solid product obtained from the coagulation of milk proteins, and it may contain fat and other components

C) **Sugar in water** - This is a solution. When sugar (solute) is added to water (solvent) and dissolves completely, it forms a homogeneous mixture called a sugar solution.

D) **Sawdust in water** - This is not a solution. Sawdust is insoluble in water, so it will not dissolve.

18 Correct option: (D)

Air is a mixture of different gases, such as nitrogen, oxygen, carbon dioxide, and trace amounts of other gases. Since all these gases are distributed well and mixed together, it forms a gas-gas solution.

19. Correct option: (C)

Solid solution → Camphor in air

The reason is that camphor in air does not form a solid solution. Camphor is a solid compound, and when it is placed in the air, it undergoes sublimation, means

it transitions directly from a solid to a gaseous state without forming a solution with the air. Therefore, camphor in air is not a solid solution.

20. Correct option: (B)

Amalgam is a type of alloy where mercury is the solvent (liquid) and another metal, such as silver, gold, or tin, is the solute (solid). The metals are dissolved in the liquid mercury, forming a homogeneous mixture, which is a liquid in a solid state.

21. Correct Answer is: 6 types

Based on the physical state of the solute and solvent present in a solution, solutions are classified into six types:
Gas in Gas (gaseous solution)
Gas in Liquid (liquid-gas solution)
Gas in Solid (solid-gas solution)
Liquid in Liquid (liquid-liquid solution)
Liquid in Solid (solid-liquid solution)
Solid in Solid (solid-solid solution)

22. Correct Answer is: 4

The number of components in a quaternary solution is four.

23. Correct Answer is: one phase.

In a true solution, there is only one phase present. A true solution is defined as a homogeneous mixture of two or more substances where the solute particles are uniformly distributed and individual solute particles cannot be observed with the naked eye or through simple filtration.

Classification of Solutions

1. Correct option: (C)

A solution that cannot dissolve any more solute at a given temperature is known as a "Saturated solution." In a saturated solution, the maximum amount of solute has been dissolved in the solvent and any additional solute added will not dissolve but will remain undissolved at the bottom of the container.

2. Correct option: (B)

A solution that can dissolve some more solute at a given temperature is known as an "**Unsaturated solution**." In an unsaturated solution, the amount of solute dissolved in the solvent is less than the maximum amount that could be dissolved at that temperature.

3. Correct option: (A)

A solution that has excess of solute dissolved in it than required for its saturation at a given temperature is known as a "Super saturated solution." That means more solute has been dissolved in the solvent than required under that specific temperature and conditions. Super saturated solutions are often formed by dissolving the solute at a higher temperature and then cooling the solution slowly without allowing the excess solute to precipitate out. As a result, the solution becomes unstable and can crystallize if disturbed, releasing the excess solute and returning to a saturated state.

4. Correct option: (A)

Milk is actually a colloidal suspension, not a true solution. It is composed of water as the solvent and various particles, including proteins, fats, and minerals, as the solute. These particles are dispersed throughout the liquid but do not dissolve completely.

5. Correct option: (D)

Cough syrup contains solid particles or droplets of active ingredients that do not dissolve completely in the liquid. These solid particles remain suspended in the liquid, and over time, they may settle down at the bottom. To use cough syrup, you usually need to shake the bottle to disperse the solid particles before taking the medicine.

6. Correct option: (A)

The phenomenon of scattering of visible light by the particles of a colloid is known as the Tyndall effect. When a beam of light passes through a colloid, it gets scattered by the dispersed particles, making the path of the beam visible. This effect is not observed in true solutions, where the particles are too small to scatter light significantly.

7. Correct option: (B)

A colloid is a type of mixture in which the size of the solute particles is within the range of approximately 1 nanometer to 100 nanometers. In colloids, the particles are larger than those found in true solutions but smaller than those in suspensions.

8. Correct option: (C)

Salt and water

The Tyndall effect is defined as scattering of light by the particles in a colloid or suspension. It is observed when the particles in the mixture are large enough to scatter light. Among the options provided, salt and water form a true solution because the salt particles dissolve completely in water, and the resulting mixture is homogeneous.

9. Correct option: (B)

The solution in which the size of the solute particles is greater than 100 nanometers is called a colloid. In colloids, the particle size falls in the range of approximately 1 nm to 1000 nm, and these particles are large enough to scatter light, resulting in the Tyndall effect.

10. Correct option: (B)

A heterogeneous mixture is defined as components are not uniformly distributed throughout the mixture. Milk is a heterogeneous mixture because it consists of various components like water, fats, proteins, vitamins, and minerals, which are not evenly dispersed throughout the liquid.

---

11. Answer: B) Heterogeneous mixtures

- *Explanation:* Suspensions contain large particles that do not mix uniformly and remain physically distinct from the liquid.

12. Answer: C) They settle down

- *Explanation:* Due to gravity and the relatively large size of the particles, suspended solids eventually settle to the bottom over time.

13. Answer: C) Suspension

- *Explanation:* Chalk powder does not dissolve in water; it forms a turbid, milky mixture where large chalk particles remain suspended temporarily before settling down.

14. Answer: C) Cheese

- *Explanation:* Cheese is a gel-type colloid (liquid dispersed in a solid medium). Salt solution, sugar solution, and copper sulphate solution are all true solutions.

LEVEL – II:

1. Answer: B) Unsaturated solution

- *Explanation:* An unsaturated solution has the capacity to dissolve more solute at a given temperature.

2. Answer: B) Saturated solution

- *Explanation:* A saturated solution contains the maximum amount of solute that can be dissolved at that specific temperature.

3. Answer: C) Supersaturated solution

- *Explanation:* A supersaturated solution holds more dissolved solute than it theoretically should at that lower temperature, making it unstable.

4. Answer: B) Particles settle down

- *Explanation:* Suspensions scatter light (Tyndall effect) while particles are floating. Once the large particles settle at the bottom, the liquid above becomes clear and stops scattering light.

5. Answer: D) Muddy water

- *Explanation:* Muddy water consists of soil and clay particles suspended in water that eventually settle out. Milk is a colloid, while salt and sugar solutions are true solutions.

6. Answer: C) Cheese

- *Explanation:* The Tyndall effect is the scattering of light by colloidal or suspended particles. Cheese is a colloid, so it scatters light. True solutions (salt and sugar in water) do not show the Tyndall effect.

7. Answer: C) True solution

- *Explanation:* True solutions have the smallest particle sizes (less than 1 nm or 10^{-9} m), which is why they do not scatter light or settle out.

8. Correct options: (A, B, C, D)

A) **A solution in which alcohol is used as solvent is called an alcoholic solution** - This statement is correct. When alcohol is used as the solvent for other solute which are dissolved in it, the resulting solution is called an alcoholic solution. For example, if you dissolve sugar in alcohol, you get an alcoholic solution of sugar.

B) **A solution that cannot dissolve any more solute at a given temperature is known as a saturated solution** - This statement is correct. A saturated solution is one where the maximum amount of solute that can be dissolved in the solvent has already been dissolved at a specific temperature. Any additional solute added will not dissolve and will remain as undissolved particles.

C) **A solution that can dissolve some more solute at a given temperature is known as an unsaturated solution** - This statement is correct. An unsaturated solution is one where more solute can be dissolved in the solvent at a specific temperature. It means the solution has not reached its maximum capacity to hold more solute.

D) **All solutions are mixtures** - This statement is correct. A solution is a type of mixture where two or more substances are mixed together, but they do not chemically react with each other. Instead, they form a homogeneous mixture where the components are uniformly distributed throughout.

9. Correct options: (B, C and D)

Except option A all the given options are incorrect.

By using process filtration and decantation we will not be able to separate all the components in every conditions of the solution. We know that solution is a homogeneous mixture, and it cannot have any particular chemical formula.

10. **Answer: (C) Assertion is true, but Reason is false.**
Explanation: The Assertion accurately describes the Tyndall effect. However, the Reason is completely false—colloidal particles are too small (between 1 nm and 100 nm) to be seen individually with the naked eye.

11. **Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**
Explanation: Particles in a suspension are quite large (exceeding 100 nm or 10^{-7} m), which is why they easily settle down, scatter light beams while suspended, and can be seen clearly without a microscope.

12. a → P; b → S; c → Q; d → R

13. **Answer: B) They are too small**
Explanation: Their individual diameter falls between 1 nm and 100 nm, which is far below the resolution limit of the human eye.

14. **Answer: C) Particles do not settle down**
Explanation: Stability in a mixture means the dispersed phase remains suspended indefinitely. Unlike suspensions, stable colloidal particles stay scattered because of constant Brownian motion and electrostatic repulsion.

15. **Answer: C) Scattering of light**
Explanation: This phenomenon is known as the Tyndall effect, where the colloidal particles deviate the incoming light rays in all directions, rendering the entire path of the beam visible to your eyes.

16. **Correct Answer is: 3 types**.
Depending on the size of the solute particles, solutions can be classified into three main types:
True Solutions
Colloidal Solutions
Suspension Solutions

17. **Correct Answer is: 4 types**.
Dilute Solutions
Concentrated Solutions
Saturated Solutions
Supersaturated Solutions

Previous year questions

1. **Correct Option: C) II and IV only**
Explanation:

- Tyndall Effect: Illuminating a path of light is known as the Tyndall effect.
- True Solutions: They do not scatter light because their particles are too small.
- Colloids: They have larger particles that scatter light and illuminate the path.
- Analysis of substances:
  - (I) Blood: It is a colloid. It scatters light.
  - (II) Brine: It is a true solution of salt in water. It does not scatter light.
  - (III) Milk: It is a colloid. It scatters light.
  - (IV) Soda water: It is a true solution of carbon dioxide in water. It does not scatter light.

Therefore, Brine (II) and Soda water (IV) will not illuminate the path of light.

2. **Correct Option: D) Both B and C.**
Explanation:

- Particulate Nature: The experiment shows that matter is not a continuous block. It is made of tiny, discrete particles.

- Intermolecular Spaces: Liquid water has empty spaces between its particles.
- Dissolution Process: When stirred, salt particles break down and fit into these empty spaces.
- Observation: This explains why the water level does not change significantly when salt dissolves.

4. Chemistry in Daily Life

Introduction and uses of some important elements

Level-1

1. Correct option: (D)

Among the options given, bromine (Br_2) is present in a liquid state at room temperature. The others fluorine (F_2), hydrogen (H_2), and chlorine (Cl_2) are gases at room temperature.

2. Correct option: (A)

Graphite is a good conductor of electricity primarily due to the presence of free electrons. In the structure of graphite, carbon atoms are arranged in hexagonal layers, and each carbon atom is bonded to three others in a trigonal planar arrangement. The fourth outer electron of each carbon atom is free to move within the layers, forming a sea of delocalized electrons. These free electrons are able to carry an electric current, making graphite a good conductor.

3. Correct option: (D)

Duralumin is an aluminium alloy that typically contains aluminium, copper, and small amounts of other elements.

4. Correct option: (C)

Alloyed gold often contains silver or copper, or a combination of both. These metals are added to gold to create alloys with desirable properties, such as increased hardness or changes in colour.

5. Correct option: (A)

Bronze is generally more resistant to corrosion than brass. Bronze is an alloy of copper and tin, and its corrosion resistance can be attributed to the formation of a protective layer of copper oxide on its surface. On the other hand, brass is an alloy of copper and zinc and may be more susceptible to corrosion, especially in certain environments.

6. Correct option: (D)

Stainless steel is an alloy that typically contains iron (Fe), chromium (Cr), nickel (Ni), and other elements. The combination of these elements imparts corrosion resistance, strength, and other desirable properties to stainless steel. Therefore, all the elements mentioned in the options iron, chromium, and nickel are commonly present in stainless steel.

Here are the step-by-step solutions and clear explanations for all the questions provided.

7. Correct Option: C) Zinc

- **Explanation:** Tensile strength measures a metal's resistance to breaking under tension. Among the given options, Iron, Copper, and Aluminium have high tensile strengths and are widely used structurally. Zinc is relatively brittle at room temperature and possesses the lowest tensile strength of the four.

8. Correct Option: D) Magnesium

- **Explanation:** An alloy is a homogeneous mixture of two or more metals (or a metal and a non-metal). Brass (Copper + Zinc), Bronze (Copper + Tin), and Magnalium (Aluminium + Magnesium) are alloys. **Magnesium** is a pure chemical element (Mg).

9. Correct Option: D) Radon

- **Explanation:** Helium, Neon, and Argon are chemically inert and stable under high temperatures, making them safe for electrical bulbs. **Radon**, however, is highly radioactive and dangerous, so it cannot be used for commercial lighting.

10. Correct Option: B) Poor electrical conductivity

- **Explanation:** Silicone (silicon-based polymers) is valued in cookware for its ability to withstand heat, resist chemical reactions, and prevent food from sticking. While it is a poor electrical conductor, this electrical property is completely irrelevant to its function on a kitchen stovetop.

11. Correct Option: D) Zinc

- **Explanation:** PCBs rely on metals with exceptionally high electrical conductivity to route signals efficiently. Copper, Silver, and Aluminium are excellent choices. **Zinc** is a poor electrical conductor relative to these metals and is prone to oxidizing rapidly, making it unsuitable for internal PCB traces.

12. Correct Option: C) It is soft in nature

- **Explanation:** Graphite has a layered crystalline structure held together by weak Van der Waals forces. This layered arrangement allows the sheets to slide over one another easily, making graphite soft and slippery. When pressed onto paper, these layers flake off easily to leave behind dark marks.

13. Correct Option: C) Silicon dioxide

- **Explanation:** Sand is primarily composed of quartz grains, which chemically consists of **Silicon dioxide** (SiO_2).

14. Correct Option: D) Copper

- **Explanation:** Telecommunication cables require high electrical conductivity and high ductility to carry signals over long distances without high resistance losses. **Copper** strikes the best balance of exceptional conductivity and durability among the options listed.

15. Correct Option: B) Tin

- **Explanation:** Tinning is the specific process of thinly coating sheets of iron or steel with a layer of **Tin** to protect them from rusting and corrosion.

Level-II:

Single Option Correct

1. Correct Option: B) Composition and behaviour of substances

- **Explanation:** Chemistry is the branch of science concerned with studying the composition, structure, properties, and reactions of all forms of matter—whether they are living, non-living, metals, or gases.

2. Correct Option: B) Malleability

- **Explanation:** * **Malleability** is the unique property that allows metals to deform under compressive stress (like hammering) into thin sheets without shattering.
  - *Note: Ductility is drawing into wires; Sonority is making sound; Conductivity is carrying heat/ electricity.*

3. Correct Option: B) Sterilising water

- **Explanation:** Chlorine is a potent disinfectant and oxidizing agent. It is universally added to drinking water and swimming pools to kill harmful bacteria and microorganisms.

4. Correct Option: B) Cancer treatment

- **Explanation:** Because Radon (Rn) is radioactive, its radiation is precisely targeted in radiotherapy to destroy malignant tumor cells in cancer patients.

5. Correct Option: B) Copper, zinc and tin

- **Explanation:** Traditional bronze is primarily an alloy of **Copper and Tin**. However, industrial variants often contain a minor percentage of Zinc alongside tin to improve casting qualities. Hence, option B represents its complex structural composition best out of the provided choices.

One or More Than One Option Correct Type

6. Correct Options: A, B, and C

- **Explanation:** * **Jewellery (A):** Extremely common due to its luster and corrosion resistance.
  - **Dentistry (B):** Biocompatible and durable for crowns and fillings.
  - **Satellites (C):** Gold leaf films reflect solar radiation and protect space equipment.
  - *Note: It is never used in fertilizers.*

7. Correct Options: A, B, and D

- **Explanation:** Metals generally possess a metallic shine (**Lustre**), can be hammered into sheets (**Malleability**), and drawn into wires (**Ductility**). They are *good* conductors, making option C incorrect.

8. Correct Options: A, C, and D

- **Explanation:** Non-metals generally lack shine (**Dull**), do not conduct electricity well (**Insulators**), and can exist in various physical states, including **Gases** (e.g., Oxygen, Nitrogen).

9. Correct Options: A, B, and C

- **Explanation:** * **Iodised salt (A):** To prevent thyroid disorders like goiter.
  - **Dressing wounds & Medicines (B & C):** Tincture of iodine is a powerful antiseptic used on wounds.
  - *Note: Iodine cannot be used as a fuel.*

10. Correct Options: A and B

- **Explanation:** Both brass and bronze are highly durable, corrosion-resistant alloys traditionally used to cast artistic **Statues** and manufacture cooking **Utensils**. They are rarely used for internal electrical switches due to cheaper, highly conductive pure copper alternatives, and are completely unrelated to fertilizers.

Assertion and Reason Type

11. Correct Option: A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion

- **Explanation:** Tin does not react with food acids easily and acts as a barrier layer over underlying steel/iron, protecting the underlying structural metal from moisture-induced rusting.

12. Correct Option: A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion

- **Explanation:** Pure metals have neatly arranged layers of atoms that slide past one another easily. Adding foreign atoms during alloying disrupts this uniformity, locking the structure in place and enhancing overall mechanical properties like tensile strength and hardness.

Match the Following

13. Correct Matches: P → b, Q → c, R → a, S → d

- **Detailed Explanation:**
  - **P. Gold** matches **b (Jewellery)**: Highly valued for its anti-reactive properties and shine.

- **Q. Copper** matches **c (Electric wires)**: Possesses excellent domestic electrical conductivity.
- **R. Aluminium** matches **a (Transmission cables)**: Lightweight and inexpensive, ideal for overhead high-voltage power lines.
- **S. Iron** matches **d (Construction)**: Essential for structural steel rods, beams, and scaffolding.

Comprehension Type

14. Correct Option: B) Lustrous

- **Explanation:** According to the text, metals are characterized by their ability to reflect light, giving them a distinct shine (lustre).

15. Correct Option: B) Ductility

- **Explanation:** The ability of a solid material to stretch and deform under tensile stress into a thin wire is defined as ductility.

16. Correct Option: B) Heat and electricity

- **Explanation:** The text explicitly notes that metals serve as excellent paths for transferring heat and electric current.

Integer Type

17. Answer: 3

- **Explanation:** Non-metals exist across all **three** primary states of matter at room temperature:
- **Solid:** Carbon, Sulfur, Phosphorus
- **Liquid:** Bromine
- **Gas:** Oxygen, Nitrogen, Hydrogen

18. Answer: -39

- **Explanation:** Mercury is the only metallic element that remains liquid at room temperature. Its exact melting point is $-38.83°C$, which rounds neatly to the integer **-39**.

19. Answer: 4

- **Explanation:** Duralumin is a strong, lightweight aluminum alloy that is made up of exactly **four** primary chemical components:

1. Aluminium (Al, $\approx 95\%$)
2. Copper (Cu, $\approx 4\%$)
3. Magnesium (Mg, $\approx 0.5\%$)
4. Manganese (Mn, $\approx 0.5\%$)

20. Answer: 2

- **Explanation:** Brass is a simple binary alloy consisting fundamentally of **two** main elements: **Copper (**Cu**)** and **Zinc (**Zn**)**.

Some Useful Compounds:

1. Answer: B) Cl_2

- Explanation: Hydrochloric acid (HCl) is commonly oxidized to prepare chlorine gas (Cl_2) in laboratory or industrial setups (for example, by reacting it with manganese dioxide: $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$).

2. Answer: D) None

- Explanation: Common salt is chemically pure Sodium Chloride (NaCl). Options A, B, and C describe mixtures with other compounds like silver chloride (AgCl) or sodium iodide (NaI), which do not constitute basic common salt.

3. Answer: D) Copper ions

- Explanation: Blue vitriol is hydrated copper sulphate ($CuSO_4 \cdot 5H_2O$). The distinctive blue color is due to the presence of Cu^{2+} (copper) ions coordinated with water molecules, which cause d-d electronic transitions that absorb red light and reflect blue light.
-

4. **Answer: D) Distillation**

- Explanation: NaCl is obtained from seawater primarily by evaporation in large shallow ponds. The crude salt is then dissolved in water and subjected to recrystallization to remove impurities. Distillation is a method used to recover pure water from a solution, not for large-scale salt harvesting.

5. **Explanation:** Vegetable oil is composed of fats. Gram for gram, fats provide the highest amount of energy (around 9 kcal/g) compared to carbohydrates (glucose, cereals) and proteins (pulses), which provide about 4 kcal/g.

6. **Answer: B) ZnO**

- Explanation: Zincite is a mineral form of zinc oxide (ZnO). Heating it strongly does not change its chemical composition, though it famously undergoes a reversible color change (turns yellow when hot and white when cold).

7. **Answer: C) Methane**

- Explanation: Biogas is primarily composed of Methane (CH_4), typically making up about 50–75% of its total volume, alongside carbon dioxide.

8. **Answer: D) Lithium iodide**

- Explanation: Iodised salt is common salt (NaCl) fortified with small amounts of potassium iodide (KI), potassium iodate (KIO_3), or sodium iodide (NaI) to prevent iodine deficiency. Lithium compounds are toxic in high doses and are never added to food.

9. **Answer: B) Carbon, hydrogen and oxyge**n

- Explanation: Starch is a complex carbohydrate (polysaccharide) with the general chemical formula $(C_6H_{10}O_5)_n$. It contains only Carbon (C), Hydrogen (H), and Oxygen (O).

10. **Answer: B) Lose their properties**

- Explanation: When elements chemically combine to form a compound, they lose their individual properties and form a new substance with entirely different characteristics (e.g., explosive sodium and toxic chlorine combine to make safe edible salt, NaCl).

11. **Answer: B) Sand and clay**

- Explanation: Extracted crude salt from natural sources contains insoluble earthy impurities like sand, clay, and other rock particles.

12. **Answer: B) Preservative**

- Explanation: Sodium chloride (NaCl) draws moisture out of food items via osmosis, preventing bacterial growth, making it an excellent traditional preservative for pickles, meat, and fish.

13. **Answer: A) Cleaning metals**

- Explanation: HCl is widely used in an industrial process called "pickling" to clean rust, scale, and oxide impurities from the surfaces of iron or steel metals.

14. **Answer: B) Copper sulphate**

- Explanation: Blue vitriol is the common name for hydrated copper(II) sulphate ($CuSO_4 \cdot 5H_2O$).

15. **Answer: B) Calcium carbonate**

- Explanation: Limestone is a sedimentary rock composed primarily of the mineral calcite, which is chemically calcium carbonate ($CaCO_3$).

Level-II

1. **Answer: A) Zinc carbonate**

- **Explanation:** Heating zinc carbonate ($ZnCO_3$) decomposes it into zinc oxide (ZnO) and carbon dioxide (CO_2) via thermal calcination: $ZnCO_3 \xrightarrow{\Delta} ZnO + CO_2$.

2. **Answer: B) $CaSO_4 \cdot 2H_2O$**

- Explanation: Gypsum is a soft sulfate mineral composed of calcium sulfate dihydrate, denoted chemically as $CaSO_4 \cdot 2H_2O$.

3. **Answer: B) Heating gypsum**

- Explanation: When gypsum ($CaSO_4 \cdot 2H_2O$) is heated to about 100°C (373 K), it loses three-fourths of its water of crystallization to become Plaster of Paris ($CaSO_4 \cdot \frac{1}{2}H_2O$).

4. **Answer: B) Hardest compound**

- Explanation: Silicon carbide (SiC), also known as carborundum, is an extremely hard, synthetically produced crystalline compound used widely as an industrial abrasive.

5. **Answer: A) CH_4**

- Explanation: Methane is the simplest alkane hydrocarbon, consisting of one carbon atom bonded to four hydrogen atoms (CH_4).

6. **Answer: A) C, H, O**

- Explanation: Carbohydrates are organic molecules made fundamentally of Carbon (C), Hydrogen (H), and Oxygen (O), usually in a 1:2:1 ratio.

One or More Than One Option Correct Type

7. **Answers: A, B, and D**

- Explanation: Fats provide the maximum energy per gram among macronutrients (9 kcal/g), are chemically composed of C, H, and O, and are stored in the body (in adipose tissues) as reserves. They do *not* contain nitrogen.

8. **Answers: A, B, and C**

- **Explanation**: Proteins contain nitrogen (in amino acids), are critical building blocks to help growth, and work actively to repair damaged tissues. They are *not* the body's only energy source.

9. **Answers: A, B, and C**

- **Explanation**: Hydrocarbons contain only carbon and hydrogen atoms. They release a large amount of energy upon combustion and are heavily utilized as fuels (e.g., LPG, CNG, petrol). They do *not* contain oxygen.

10. **Answers: A, B, and C**

- **Explanation:** Silicon dioxide (SiO_2) is the chief constituent of sand, and it is widely used as a raw material in manufacturing both glass and cement. It is a solid polymer network, not a gas.

Assertion and Reason Type

11. **Answer: B)** Both Assertion and Reason are true, but Reason is NOT the correct explanation.

- **Explanation**: Both statements are factually true. Limestone ($CaCO_3$) is a vital raw material for cement. However, simply stating its chemical identity ($CaCO_3$) does not logically explain *why* it is used in cement production (it is used because thermal decomposition yields calcium oxide (CaO), which reacts with silicates to form the binding component of cement).

12. **Answer: B)** Both Assertion and Reason are true, but Reason is NOT the correct explanation.

- Explanation: Both statements are correct. Proteins are vital for cell growth/repair, and they do contain nitrogen. However, containing nitrogen isn't the direct reason *why* they repair tissues (tissue repair happens because they form complex structural chains, enzymes, and hormones).

Match the Following

13. P → b: Sodium chloride is used as a Preservative.

- Q → a: Hydrochloric acid is used for Cleaning metals.
- R → d: Blue vitriol finds applications as an Insecticide/Fungicide (e.g., Bordeaux mixture).
- S → c: Limestone is a core component in Cement manufacture.

Comprehension Type

14. **Answer: B) Recrystallisation**

- Explanation: As noted in the passage text, crude sea salt contains soluble and insoluble impurities. Dissolving it and letting it cool/evaporate systematically allows pure NaCl crystals to separate out via recrystallization.

15. **Answer: A) Cooking**

- Explanation: Sodium chloride (NaCl) is our essential daily table salt used globally in cooking meals.

16. **Answer: A) Solute from solvent**

- Explanation: Recrystallization works by dissolving a crystalline solid mixture in a minimum amount of hot solvent and letting the desired solute crystalize out cleanly as it cools down, leaving the impurities behind in the solvent.

Integer Type

17. Number of oxygen atoms in one molecule of sodium chloride = 0

- Explanation: The chemical formula for sodium chloride is NaCl. It contains only one sodium atom (Na) and one chlorine atom (Cl). There are no oxygen atoms.

18. Number of elements present in calcium carbonate ($CaCO_3$) = 3

- Explanation: The elements present in $CaCO_3$ are Calcium (Ca), Carbon (C), and Oxygen (O). Thus, there are exactly 3 unique elements present.

Some Important Industrially Useful Materials

LEVEL 1

1. **Correct option: (D)**

Crook's glass (cerium oxide) which absorbs high UV light and is used to prepare spectacles.

2. **Correct option: (D)**

The basic components of paint include:

(a) Pigment: Provides colour and opacity to the paint.

(b) Vehicle (or binder): Carries the pigment and binds it to the painted surface.

(c) Solvent: A liquid that helps to adjust the consistency of the paint and evaporates after application.

3. **Correct option: (C)**
Vehicle tires are typically made of rubber, which is a polymer. Rubber is composed of long chains of repeating molecular units, making it a polymer. The other options—cement, glucose, and common salt—do not have polymers in their basic composition.

4. **Correct option: (A)**
Calcium oxide, represented as CaO, is commonly known as lime. It is produced by heating calcium carbonate ($CaCO_3$) to high temperatures, causing it to undergo thermal decomposition and release carbon dioxide, leaving behind calcium oxide. Lime is used in various applications, including as a construction material and in agriculture.

5. **Correct option: (A)**
Portland cement is typically made by combining limestone and clay in specific proportions and then heating the mixture in a kiln. The process of heating the mixture results in the formation of clinker, which is then ground into a fine powder to produce Portland cement. The correct option is the mixture of limestone and clay.

6. **Correct option: (C)**
In the dry process of cement manufacturing, the fine powder of limestone and clay is produced through the grinding process. The raw materials are first quarried and then crushed into smaller pieces before being finely ground into the powder form. Therefore, the correct term for this process is "grinding."

7. **Correct option: (A)**
Silica, also known as silicon dioxide (SiO_2), is a key component used in the making of glass. Silica is often derived from sand and provides the main structural component of glass. Other materials, such as soda ash and limestone, may also be added to the mix depending on the type of glass being produced.

8. **Correct option: (A)**
Feldspar is used in the making of glass. It is a common raw material in glass manufacturing, contributing to the composition of the glass batch along with silica (sand) and other materials. While silicone is related to silicon, it is not typically used directly as a primary raw material in glass making.

9. **Correct option: (A)**
The blowing of glass is commonly used in the manufacture of glassware. This technique involves shaping molten glass by blowing air into a gather of glass on the end of a blowpipe, allowing artisans to create various glass objects such as bottles, vases, and other decorative or functional items.

10. **Correct option: (C)**
A monomer is the fundamental unit or building block of a polymer. Monomers are small molecules that can undergo polymerization, a chemical reaction where they link together to form a larger molecule called a polymer. The process of combining monomers to form a polymer involves repeated chemical units, resulting in a macromolecule with a high molecular weight.

11. **Correct option: (B)**
The monomer of polyethylene is ethylene. Polyethylene is a polymer made by polymerizing ethylene molecules, which consist of two carbon atoms and four

hydrogen atoms. The chemical formula of ethylene is C_2H_4 and through polymerization, it forms the repeating units of polyethylene.

12. Correct option: (D)

"Chain" is not a polymer; it is a term that can refer to a structure or arrangement but is not a specific polymer material. PVC (Polyvinyl chloride), plastic (which is a general term for various synthetic or semi-synthetic polymers), and Nylon are examples of polymers.

13. Correct option: (C)

Reusable plastics are often referred to as thermoplastics. Thermoplastics are a type of polymer that becomes mouldable when heated and solidifies upon cooling. This property allows them to be melted and reshaped multiple times, making them suitable for reuse. Examples of thermoplastics include polyethylene, polypropylene, and polyvinyl chloride (PVC).

14. Correct option: (D)

All the options—melamine, formaldehyde, and Bakelite—are examples of thermosetting plastics. Thermosetting plastics are polymers that, once moulded and set, cannot be melted or reshaped through heating. They undergo a chemical change during the curing or setting process, creating a strong and rigid structure. Melamine, formaldehyde (used in the production of phenolic resins), and Bakelite are common examples of thermosetting plastics.

15. Correct option: (C)

Starch is a carbohydrate. Carbohydrates are organic compounds made up of carbon, hydrogen, and oxygen, and they serve as a primary source of energy for living organisms. Starch is a complex carbohydrate consisting of many glucose units linked together. Lemon, butter, and salt, on the other hand, do not belong to the carbohydrate group.

16. Correct option: (D)

Keratin is a fibrous structural protein that is a major component in the structure of hair, nails, and the outer layer of the skin. It belongs to the protein category.

17. Correct option: (A)

Toilet soap typically contains better quality and more refined vegetable oils or fats compared to laundry soap. The quality of vegetable oil used in soap production affects the soap's texture, lather, and overall quality. Toilet soaps are designed for personal hygiene and often contain higher-quality ingredients to be gentler on the skin compared to laundry soaps, which are intended for cleaning fabrics.

LEVEL – II

1. Correct option: (A)

The chemical composition of limestone is primarily calcium carbonate ($CaCO_3$). Limestone is a sedimentary rock composed mostly of the mineral calcite, which is a crystalline form of calcium carbonate.

2. Correct option: (A)

Cement mortar is commonly used for plastering. Plastering is the process of covering walls and ceilings with a layer of plaster, and cement mortar, which is a mixture of cement, sand, and water, is often employed for this purpose. While concrete and reinforced cement concrete (RCC) are used for structural purposes, they are not typically used directly for plastering.

3. Correct option: (D)

All the statements are correct:

Thermoplastics can be processed repeatedly because they soften when heated and harden when cooled without undergoing a chemical change.

Thermosetting plastics, on the other hand, cannot be softened by heating after they have undergone the curing or setting process, as they undergo a chemical change during this process.

Thermoplastics do not undergo a chemical change on heating; they retain their original chemical structure.

4. Correct option: (D)

Sodium lauryl sulphate is a detergent. It is a surfactant commonly used in various cleaning and personal care products due to its ability to lower the surface tension of water and enhance the cleaning properties of the product.

sodium stearate, sodium chloride, and sodium carbonate are not typically used as detergents. Sodium stearate is a soap, sodium chloride is table salt, and sodium carbonate is washing soda.

5. Correct option: (C)

Mortar is a mixture used in masonry construction to bind building blocks or stones. It typically consists of cement, sand, and water. The cement binds the sand particles together when it hydrates, creating a strong and durable bond between the masonry units. The proportion of each component can vary depending on the specific application.

6. Correct option: (D)

Limestone is used in the manufacture of quicklime (calcium oxide), glass, and cement. The specific use of limestone depends on the process it undergoes:

Quicklime (Calcium oxide) is produced by heating limestone, and it is used in various applications, including as a flux in the production of iron and steel.

Limestone is a key raw material in the production of glass.

In the cement manufacturing process, limestone is one of the main raw materials used to produce clinker, which is then ground to make cement.

7. Correct option: (C)

Potassium hydroxide (KOH) is used as one of the raw materials for making soft soap. The process involves saponification, where potassium hydroxide reacts with fats or oils to produce soap and glycerol. Soft soap is a soap product that is typically made with potassium hydroxide instead of sodium hydroxide, which is used in the production of hard soaps.

8. Correct option: (A)

The correct sequence for the manufacture of glass is:

Mixing of raw materials in a proper proportion (2)

Addition of cullet (1)

Addition of colouring pigments (3)

Annealing (4)

9. Correct option: (D)

The correct sequence for the manufacture of common salt is:

Evaporation of seawater (2)

Dissolution in water (4)

Addition of sodium iodide and potassium iodide (3)

Recrystallization (1)

10. Correct option: (D)
Argon gas is heavier than air, so it cannot be used to fill weather observation balloons, as it does not allow the balloons to reach greater heights.

11. Correct option: (D)
The correct sequence for the manufacture of sodium chloride from seawater is as follows:
(4) Collection of seawater
(2) Evaporation
(1) Making of a hot saturated solution
(3) Recrystallisation

One or More Than One Option Correct Type

12. Correct Options: A, B, and C

- Explanation: * Thermoplastics have linear or branched polymer chains with weak intermolecular forces. When heated, they melt (a physical change, not chemical) and can be remolded repeatedly (A and C are correct).
- Thermosetting plastics form cross-linked chemical bonds upon heating, setting permanently into a rigid structure. They cannot be remelted or softened by reheating (B is correct).

13. Correct Options: A and B

- Explanation: Lime mortar is a paste made by mixing slaked lime, sand, and water. It has been used for centuries in masonry to bind or join bricks together and to provide a smooth finish by plastering walls.

14. Correct Options: A, B, and D

- Explanation: The primary raw materials required to manufacture Portland cement are Limestone (calcium carbonate) and Clay (silica, alumina, and iron oxide). Gypsum ($CaSO_4 \cdot 2H_2O$) is added in small amounts during the final grinding process to control the setting time of the cement.

15. Correct Options: A and D

- Explanation: Bricks are primarily manufactured from clay (earth rich in silica and alumina), which is mixed with water to form a plastic paste suitable for molding into shapes before drying and burning in a kiln.

16. Correct Options: A and C

- Explanation: Soaps and detergents are surfactants (surface-active agents) that lower the surface tension of water. They possess a hydrophobic tail that attaches to grease/oil and a hydrophilic head that dissolves in water. This allows them to trap oil droplets and lift them into the water via emulsification.

17. Correct Option: A

- Explanation: Quinine is a natural antimalarial drug extracted from the bark of the Cinchona tree. *Note: Penicillin is derived from a fungus (not a plant), Sulphadiazine is fully synthetic, and Methane is an organic gas.*

Assertion and Reason Type

18. Correct Option: (C) A is true, R is false

- **Explanation:** The assertion is true; thermoplastics can be remelted repeatedly. However, the reason is false because melting a thermoplastic is a purely physical change (breaking weak intermolecular bonds), not a chemical change.

19. **Correct Option: (A)** Both A and R are true, and R is the correct explanation of A

- **Explanation**: Soaps act as surfactants by orienting their hydrophobic tails toward the grease and hydrophilic heads toward water, trapping the grease inside spherical structures called micelles. This stabilizes the oil-in-water emulsion, making it easy to wash away.

Match the Following

20. Correct Matches: P → c, Q → a, R → d, S → b

- Detailed Explanation:
  - P. Calcined Lime matches c: Calcination involves heating limestone ($CaCO_3$) to high temperatures (≈ 1000°C) to produce quicklime (CaO).
  - Q. Cement matches a: Manufactured primarily by heating a mixture of Limestone and Clay.
  - R. Brick matches d: Formed by mixing Clay and Water, then molding and baking.
  - S. Glass matches b: Produced by fusing a mixture of Silica (sand), Limestone, Feldspar, and Cullet (recycled broken glass).

Comprehension Type

21. **Correct Option: A)** Combination of monomers

- Explanation: As stated in the passage, polymers are macromolecules built by joining thousands of small repeating units called monomers together.

22. **Correct Option: A)** Be remelted and reused

- Explanation: According to the text, thermoplastics have the property to be softened by heat and reshaped multiple times.

23. Thermosetting plastics:

- Correct Option: A) Cannot be softened by heating
- Explanation: The text explicitly mentions that thermosetting plastics change permanently upon heating and cannot be softened or remolded.

Integer Type

24. **Answer: 48**

- **Explanation:**

$$\text{Ratio} = \frac{\text{Weight of Cement}}{\text{Weight of Sand}} = \frac{1}{4}$$

$$\frac{12}{\text{Weight of Sand}} = \frac{1}{4} \Longrightarrow \text{Weight of Sand} = 12 \times 4 = 48 \text{ kg}$$

25. **Answer: 44**

- Explanation: The balanced chemical equation for the combustion of methane is: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

1. Calculate the molar mass of CH_4: $12 + 4(1) = 16$ g/mol
2. Calculate the molar mass of CO_2: $12 + 2(16) = 44$ g/mol
   From the stoichiometry, 1 mole of CH_4 (16 g) produces exactly 1 mole of CO_2 (44 g).

5. Air and Oxygen

LEVEL I

1. Correct option: (B) Decreases

As altitude increases, there are fewer air molecules above, resulting in reduced pressure exerted on the surface.

2. Correct option: (B) troposphere

In the troposphere, the temperature generally decreases with altitude. The reason is the troposphere's gases absorb very little of the incoming solar radiation. Instead, the ground absorbs this radiation and then heats the tropospheric air by conduction and convection.

3. Correct option: (C) sulphur dioxide

Suphur dioxide (SO_2) is not a greenhouse gas, thus it does not contribute to global warming.

4. Correct option: (C) O2

Oxygen is the most abundant element in the earth crust. Silicon is the second most abundant element and aluminium is the 3rd.

5. Correct option: (A) deep-sea diving

In the sea, pressure is more, so more amount of oxygen
gets dissolved in the blood. This excess oxygen when dissolved in the blood causes several health problems like nausea, dizziness, etc.

6. Correct option: (D) formation of auroras

Formation of auroras involves collision of electrons with molecules of N2 ana O2 and emission of beautiful radiations. It does not result in formation of nitrates.

7. Correct option: (D) 45312

Explanation: Decreasing order means highest to lowest altitude: Exosphere (4) → Ionosphere/Thermosphere (5) → Mesosphere (3) → Stratosphere (1) → Troposphere (2).

8. Answer: C) O2

Explanation: Pure oxygen makes combustion highly rapid and dangerous. Atmospheric nitrogen (78%) dilutes oxygen (21%), controlling the rate of burning and respiration.

9. Correct option: (C) 21534

Inversion layer:

- The layer above the troposphere is called the stratosphere and it extends up to 50 km above the earth's surface. Above 25 km, temperature increases due to ultraviolet (UV) radiation absorption. This region of increase in temperature is called the inversion layer.

Convection currents:

- The layer of the atmosphere which extends up to 20 km above the earth's surface is called the troposphere. Convection currents are present in this layer.

Auroras:

- The layer of the atmosphere which extends up to 690-950 km above the earth's surface is called the thermosphere or ionosphere.
- Auroras appear in this layer. Auroras are the luminous streams that are visible from the earth's surface at northern and southern latitudes.

Harmful ionizing radiations:

- The layer of the atmosphere which extends up to 10,000 km above the earth's surface is called the exosphere. This layer absorbs harmful ionizing radiation.

Burning up of meteorites:

- The layer of the atmosphere which extends up to 85 km above the earth's surface is called the mesosphere. Meteorites burn up in this layer.

10. Correct option: (B) 3241

Step 1: In the first step, nitrogen combines with oxygen to form Nitric oxide
Step 2: Nitric oxide further reacts with oxygen to form Nitrogen dioxide
Step 3: Now, Nitrogen dioxide combines with water and oxygen to form Nitric acid.
Step 4: In the last step, Nitric acid reacts with metal carbonates to form corresponding metal nitrate, Carbon dioxide and water.

11. Answer: C) Nitrogen (78%)

12. Answer: C) Oxygen

13. Answer: A) Lighter

Explanation: Heating causes air to expand, decreasing its density and making it rise.

14. Answer: D) 13 km

15. Answer: B) decreases

Explanation: Atmospheric pressure always decreases with altitude because the mass of the air column above decreases.

16. Answer: A) Ozone

Level-II

1. Answer: D) All of These

2. Answer: A) Temperature

The water boils at

3. Answer: D) 100°C (at standard atmospheric pressure)

4. Answer: B) Sea Level

Explanation: Air density and weight are highest at the lowest surface altitude.

5. Answer: A) Loo (a dusty, hot, summer wind in South Asia)

6. Answer: A) Sea breeze

7. Answer: C) Precipitation

8. Answer: B) atmosphere

9. Answer: C) oxygen

10. Answer: A) Carbon dioxide (a primary greenhouse gas)

11. Answer: A) Thermosphere

12. Answer: B) rain gauge

13. Answer: D) Temperature

Explanation: Insolation is incoming solar radiation, which directly governs global temperatures.

One or more than one option correct type

14. Answer: A), B), C), and D)

Explanation*:* Oxygen gas is completely colorless, odorless, tasteless, and slightly heavier than air.

15. Answer: (A), (B), and (D)

Explanation*:* Plants cannot directly absorb gaseous nitrogen via leaves (Option C); they rely on soil fixation or chemical fertilizers.

16. **Answer: (A), (B), (C), and (D)**
***Explanation**:* While (B) is the most common modern school lab method, all four chemical pathways successfully generate oxygen gas in laboratory or industrial frameworks.

17. **Answer: (A) and (B)**
***Explanation**:* Oxygen is *not* combustible (it does not burn on its own; it supports combustion, making C false), and it is neutral to litmus but chemically active (making D false).

Assertion and reason type

18. **Answer: C)** A is correct, but R is incorrect
Explanation*:* Oxygen gas itself is neutral (A is true). However, Reason (R) is false because oxygen reacts with non-metals too (forming acidic oxides like CO_2 or SO_2).

19. **Answer: C)** A is correct, but R is incorrect
***Explanation**:* Nitrogen is used to flush food packaging (like potato chips) because it is chemically unreactive (inert), which prevents oxidation and spoilage. Thus, the reason statement is factually incorrect.

20. **Answer: P → c, Q → a, R → b, S → d**

Comprehension type

21. **Answer: B)** Stratosphere

22. **Answer: B)** UV radiation

23. **Answer: B)** It protects life from harmful radiation

Integer type

24. **Answer: 13** (Acceptable range depending on text: **8 to 18**; average is **13**)

25. Answer: 1 (More precisely **0.93** or **0.934**)

Carbon Dioxide and hydrogen, Air pollution

LEVEL – I

1. **Answer: B)** magnesium

- Explanation: Magnesium is a highly reactive metal. It can forcefully pull oxygen away from carbon dioxide to keep burning, forming magnesium oxide and carbon deposits:
  $2Mg + CO_2 \rightarrow 2MgO + C$

2. **Answer: B) CO_2**

- Explanation: Soda water is manufactured by dissolving carbon dioxide gas in water under high pressure to create carbonation.

3. **Answer: B) dry ice**

- Explanation: Dry ice (solid carbon dioxide) has a very low temperature (−78.5°C) and sublimates directly into gas without leaving liquid residue, making it ideal for shipping seafood.

4. **Answer: D)** CO_2 water, and energy

- Explanation: Biological respiration breaks down glucose in cells to yield carbon dioxide, water, and usable chemical energy (ATP).

5. **Answer: B)** global warming

- Explanation: $CO_2$is a major greenhouse gas that traps outgoing infrared radiation in the Earth's atmosphere, driving up global temperatures.

6. **Answer: A)** alloying

- Explanation: An alloy is defined structurally as a homogeneous solid mixture consisting of two or more metals, or metals mixed with non-metals.

7. **Answer: C) H_2O_2**

- Explanation: Hydrogen peroxide (H_2O_2) easily decomposes at room temperature in the presence of a catalyst like manganese dioxide (MnO_2) to produce oxygen gas and water.

8. **Answer: B) liquid oxygen**

- Explanation: Since space lacks an atmosphere, rockets must carry their own oxidizer. Liquid oxygen (LOX) serves as the primary supporter of combustion for the rocket fuel.

9. **Answer: B) ozone**

- Explanation: Ground-level (tropospheric) ozone is toxic and forms harmful smog. Stratospheric ozone, however, is vital because it absorbs lethal solar ultraviolet (UV) radiation.

10. **Answer: B)** downward displacement of water and upward displacement of air, respectively

- Explanation: Oxygen is barely soluble in water, so it is collected over water. Carbon dioxide is heavier than air and dissolves easily in water, so it is collected by letting it sink into a container, displacing the lighter air upward.

11. **Answer: D)** nitrogen dioxide

- Explanation: While CO_2, CH_4(methane), and water vapour are major greenhouse gases, $NO_2$is an atmospheric pollutant/smog component rather than a direct greenhouse gas (unlike nitrous oxide, N_2O).

12. **Answer: A)** Na

- Explanation: Sodium (Na) imparts a distinct, bright golden-yellow color during flame tests and combustion.

13. **Answer: B) CO_2**

- Explanation: Increasing emissions of $CO_2$from fossil fuels heavily accelerate the greenhouse effect.

14. **Answer: C)** iron

- Explanation: The reaction between red-hot iron and steam is a classic reversible reaction:
  $3Fe + 4H_2O(g) \rightleftharpoons Fe_3O_4 + 4H_2(g)$

15. **Answer: A) neon**

- Explanation: Neon gas glows with a brilliant reddish-orange light when an electric discharge passes through it, which popularized "neon signs".

16. **Answer: B) Pascal**

- Explanation: The Pascal (Pa), equivalent to 1 N/m^2, is the standard SI unit for pressure.

17. **Answer: C) carbon dioxide**

- Explanation: Historically, carbon dioxide was referred to as "aerial acid" or "fixed air" because it forms acidic solutions (carbonic acid) when dissolved in water.

18. **Answer: B) stratosphere**

- Explanation: The ozone layer sits primarily in the stratosphere, roughly 15 to 35 km above Earth.

19. **Answer: D) chromium**

- Explanation: Chromium is mixed directly into iron to manufacture stainless steel, forming a passive protective layer that stops rust. *(Note: Zinc is used to coat iron via galvanization, but it is not typically an internal alloying agent for iron).*

20. **Answer: A)** low atmospheric pressure

- Explanation: As altitude increases, external atmospheric pressure drops. The air trapped inside the pen's ink reservoir remains at a higher pressure, pushing the ink outward.

21. **Answer: B)** air conditioners

- Explanation: Older cooling units and air conditioners released chlorofluorocarbons (CFCs), which break down ozone molecules in the stratosphere.

22. **Answer: D)** death of aquatic animals

- Explanation: Acid rain drastically drops the pH level of lakes and rivers, making them toxic to fish and other marine life.

23. **Answer: D)** W

- Explanation: Active metals like Na, Mg, and Kcan react with nitrogen at high temperatures to form nitrides. Tungsten (W) is highly unreactive with nitrogen, which is why it is used safely as a lightbulb filament under inert nitrogen atmospheres.

24. **Answer: C)** It is heavier than air and is highly soluble in water.

- Explanation: Collection by *upward displacement of air* means the gas settles at the bottom because it is heavier than air. It cannot be collected over water because it dissolves instantly (highly soluble).

LEVEL – II

1. **Answer: C)** It is only the human beings who need energy.

- Explanation: All living organisms (plants, animals, bacteria) require energy to perform metabolic processes and survive.

2. **Answer: A)** Carbon dioxide

3. **Answer: D)** All of these

4. **Answer: B)** Carbon dioxide

5. **Answer: B)** Jan Baptista van Helmont

- Explanation: Van Helmont recognized a distinct gas given off by burning wood and accumulating in mines around 1640, naming it *gas sylvestre*. Joseph Black later isolated and thoroughly analyzed it as "fixed air".

6. **Answer: C) Zinc**

- Explanation: Granulated zinc reacting with dilute hydrochloric acid (HCl) or sulfuric acid (H_2SO_4) is the standard method for producing hydrogen in school labs.

7. **Answer: D)** Circulatory system

- Explanation: COtargets the circulatory system by combining with hemoglobin in red blood cells to block oxygen delivery.

8. **Answer: B) Solid CO_2**

One or more than one answer type

9. **Answer: A, B, and C**

- **Explanation**: CO_2turns lime water milky by creating insoluble $CaCO_3$(A). Adding excess CO_2converts it into soluble calcium bicarbonate $\left[Ca(HCO_3)_2\right]$, dissolving the cloudiness (B and C).

10. **Answer: A, B, and C**

- Explanation: Hydrogen readily forms water with oxygen, ammonia with nitrogen, and hydrochloric acid with chlorine. It does not react with noble gases under standard conditions.

11. **Answer: A, B, and C**

- Explanation: Hydrogen is produced by reacting active metals with acids, certain metals with water/steam, or via the electrical splitting (electrolysis) of water.

12. **Answer: A, B, and C**

13. **Answer: A, B, and C**

- Explanation: Volcanic eruptions represent a major natural cause of pollution, while fossil fuels and deforestation are significant human causes. Photosynthesis removes pollution by soaking up CO_2.

Assertion and reason type

14. **Answer: A)** Both Assertion and Reason are true, and Reason is the correct explanation

- Explanation: Carbon monoxide forms carboxyhemoglobin, which binds roughly 200–250 times more tightly than oxygen, suffocating body tissues.

15. **Answer: A)** Both Assertion and Reason are true, and Reason is the correct explanation

- Explanation: Nitrogen shares a incredibly strong triple bond (N ≡ N) that requires industrial conditions (~ 450°C and 200 atmwith an iron catalyst) to combine with hydrogen.

Matching type

16. Answer: * (P) →(b) Jan Baptista van Helmont →Called CO_2"carbonium" / *gas sylvestre*.

- (Q) →(c) Joseph Black →Named CO_2"fixed air".
- (R) →(a) Antoine Lavoisier →Named hydrogen.
- (Fourth entry) →(d) Henry Cavendish →Established hydrogen as an element.

Comprehension type

17. Answer: A) CO

18. Answer: B) Acid rain

19. Answer: C) Respiratory system

Integer type

20. Vapor density of CO_2= 22

- Explanation:
  Molecular Mass of $CO_2 = 12 + (2 \times 16) = 44$ g/mol
  $\text{Vapor Density} = \frac{\text{Molecular Mass}}{2} = \frac{44}{2} = 22$

21. Explanation: Carbon monoxide possesses an affinity for human hemoglobin that is roughly 200 to 250 times greater than that of oxygen, making it highly toxic even in trace amounts.

Previous Year Questions:

1. D. Nitrogen is a gas, but it does not support burning. It is also not released during respiration, so it corresponds to S.

2. C. Layer X (Mesosphere) is actually the coldest layer of the atmosphere, while Layer W (Thermosphere) is the hottest.

3. B. Planting trees, using cleaner fuels like LPG/CNG, and checking vehicle emissions all help reduce air pollution. Treating sewage before releasing it into water mainly controls water pollution, not air pollution.

4. D. From the graph, Z ≈ nitrogen (78%), X ≈ oxygen (21%), and Y ≈ carbon dioxide (very little). Plants use carbon dioxide for photosynthesis, not nitrogen, and oxygen is not a fertiliser component, so D is incorrect.

6. Fiber to Fabric

LEVEL – I

1. Correct option: (A)

Natural fibers are threads obtained from plants and animals. Examples are cotton, coconut, husk fibers etc.

2. Correct option: (C)

The fibers obtained from plants and animals are thread like strands which are used to produce fabrics in general used in paper producing factories, cloths producing factories.

3. Correct option: (D)

Natural fibers are used to make a variety of products, including boxes (using materials like straw or bamboo), chairs (using materials like rattan or wicker) in home decoration, and clothes preparation (using materials like cotton, linen, wool, silk, etc.).

4. Correct option: (D)

Fibers are derived from plants and animals which are long thread like structures used to prepare various products. All of the options are examples of plant fibers. These natural plant fibers are commonly used in various industries, such as textile, paper, and packaging, due to their strength, durability, and other desirable properties.

5. Correct option: (C)

Among the given options plastic is not a plant fiber. It is a synthetic material produced from petrochemicals and is not obtained from any plant source. The other options (Flax, Jute, and Cotton) are all examples of plant fibers.

6. Correct option: (C)

Silk and Wool are animal fibers among the given options. Silk is produced by silkworms while Wool is obtained from the fleece of sheep and some other animals like goats (Cashmere and Mohair). These animal fibers are widely used in the textile industry for making various types of winter clothing. Remaining options are plant fibers.

7. Correct option: (B)

Buffalo does not give wool. Wool is mainly obtained from sheep, whereas other animals like goats (Cashmere and Mohair), yaks also provide certain types of wool. However, buffalo does not produce wool.

8. Correct option: (B)

Silk comes from silkworms. Silkworms are the larvae or caterpillars of the silk moth (Bombyx mori). They produce silk by spinning a cocoon made of silk fibers to protect themselves during the pupal stage of their life cycle. The silk fibers produced by the silkworms are used to make silk fabric, which is highly prized for its smooth texture and luxurious feel.

9. Correct option: (A)

Woolen fabrics have a greater bulk than cotton fabrics. Wool fibers are crimped and have a more three-dimensional structure compared to the flat and smooth structure of cotton fibers. This crimped structure gives wool bulkiness, which helps in trapping air and providing insulation in colder temperatures. Cotton fibers, being smoother and flatter, have less bulk and are more lightweight compared to wool.

10. Correct option: (A)

Woolen fabrics are known for their excellent insulation properties, making them ideal for keeping us warm during cold days. The crimped structure of wool fibers creates air pockets that trap heat, providing better insulation and retaining body warmth. During cold weather, wearing woolen clothing can help keep us cozy and comfortable.

11. Correct option: (D)

Wool comes from sheep. Sheep are domesticated animals that are raised for their wool, which is the fibrous coat that grows on their bodies. The process of obtaining wool from sheep involves shearing, where the wool is carefully cut off the sheep's body to produce various woolen products.

12. Correct option: (C)

Wool is obtained from the fleece (hair) of sheep, which is the natural covering of the sheep's body. The hair consists of wool fibers, and it is sheared off the sheep's body to yield the wool.

13. Correct option: (D)

Sheep: Sheep are the most common source of wool and produce a variety of wool types, depending on the sheep breed.

Goat: Goats provide several types of wool, including Mohair (from Angora goats) and Cashmere (Kashmir goats).

Yak: Yaks, which live in the Himalayan region, also produce a type of wool known as yak wool.

So, all three animals give wool.

14. Correct option: (B)

Air is a poor conductor of heat. Meaning it does not conduct heat well compared to materials like metals. Air has low thermal conductivity, which is why it is used as an insulating material in various applications. When there is a temperature difference between two objects, air acts as a barrier, reducing the transfer of heat between them.

15. Correct option: (D)

The Lohi breed of sheep is found in both Rajasthan and Punjab states in India. These sheep are adapted to the arid and semi-arid regions of these states and are known for their fine quality wool, which is used in the production of textiles and clothing.

16. Correct option: (D)

The Rampur Bushair breed of sheep are found in both Uttar Pradesh and Himachal Pradesh states of India. These states are known for their diverse sheep breeds, and Rampur Bushair is one of the popular sheep breeds found in these regions.

17. Correct option: (D)

The Nali breed of sheep is found in multiple states, including Rajasthan, Haryana, and Punjab. It is a popular breed known for its good wool quality and adaptability to the arid and semi-arid regions of these states. The Nali sheep are well-suited to the local climatic conditions and are reared for their wool.

18. Correct option: (A)

The Marwari breed of sheep is found in the state of Gujarat, India. It is a breed that is well-adapted to the arid and semi-arid regions of Gujarat and is known for its ability to thrive in such harsh environments. Marwari sheep are primarily reared

for their fine quality of wool, which is of good quality and suitable for various textile applications.

19. Correct option: (A)

The Patanwadi breed of sheep is found in the state of Gujarat (coastal plains of the Saurashtra and Kutch) in India. It is a popular Patanwadi sheep are primarily raised for their wool and are well-suited to the semi-arid and arid environments of Gujarat.

20. Correct option: (D)

The Rampur Bushair breed of sheep has brown fleece wool. This breed is found in Uttar Pradesh and Himachal Pradesh states of India. The wool from Rampur Bushair sheep can vary in color and is known for its good quality and unique brown hue.

21. Correct option: (C)

Wool obtained from sheep and most other wool-producing animals. The natural color of wool is generally creamy white. However, it's important to note that wool can come in various natural colors, including shades of brown, gray, and black, depending on the breed of the sheep or the animal it comes from.

22. Correct option: (C)

Yak wool is commonly found in the hilly regions of both Tibet and Ladakh. Yaks are primarily found in the high-altitude regions of the Himalayas, including parts of Tibet and the Ladakh region of India. The wool obtained from yaks, known as yak wool, is warm, durable, and suitable for making clothing and textiles that provide insulation in cold and harsh climates.

23. Answer: A) Lohi

- Explanation: Lohi is a well-known Indian breed of sheep found in Rajasthan and Punjab that is highly valued for producing excellent quality wool. *Rampur bushair* yields brown fleece, *Nali* is used for carpet wool, and *Marwari* yields coarse wool.

24. **Answer: D)** Carding machine

- Explanation: Carding is the mechanical process that untangles, cleans, and intermixes loose wool fibers to transform them into a continuous web or flat sheet ready for spinning.

25. **Answer: B)** Shearing, scouring, sorting and dying

- Explanation: The standard sequence of processing wool is:
  1. Shearing: Shaving off the fleece.
  2. Scouring: Washing the sheared hair to remove grease, dust, and dirt.
  3. Sorting: Separating hairs of different textures.
  4. Dyeing: Coloring the fibers, as natural fleece is usually black, brown, or white.

26. **Answer: D)** To prevent infection

- Explanation: Shearing involves using clippers very close to the sheep's skin, which can sometimes cause minor nicks or cuts. Dipping them in an antiseptic solution protects them from contracting bacterial or skin infections.

27. **Answer: C)** Because fleece is body hair containing germ and grease

- Explanation: Sheep fleece naturally contains significant amounts of sweat, grease (lanolin), dirt, and environmental debris trapped in the animal's hair. Cotton, being a plant fiber, requires different cleaning processes (like ginning and boiling) rather than scouring to remove heavy animal grease.

28. **Answer: D)** Woollen yarn is made up of protein, but the cotton yarn is made of cellulose.

- Explanation: Wool is composed of keratin (a protein), which has highly resilient, elastic molecular chains with strong disulfide cross-links. Cotton is composed of cellulose (a carbohydrate). The complex protein structure of wool gives it superior elasticity and tensile strength compared to individual cotton fibers.

29. **Answer: B)** Two sets of yarn are used but in knitting only one

- Explanation: Weaving involves interlacing two distinct sets of yarn (warp and weft) at right angles to create fabric. Knitting, on the other hand, creates fabric by interlocking consecutive loops using a single continuous strand of yarn.

30. **Answer: C)** Woolly dog

- Explanation: While camels, yaks, and specific breeds of goats (like Kashmir goats for Cashmere wool) yield commercial wool, domestic dogs—even fluffy "woolly" ones—are not used as sources for textile wool production.

31. **Answer: C)** Keratin

- Explanation: Just like human hair and nails, animal fleece/wool is structurally composed of the tough, fibrous protein called keratin.

32. **Answer: B)** Mulberry

- Explanation: The caterpillar of the silk moth (*Bombyx mori*) feeds exclusively on the fresh leaves of the mulberry tree to grow and produce silk.

33. **Answer: C)** Sericulture

- Explanation: Sericulture is the commercial rearing of silkworms for the production of silk. *Apiculture* is bee-keeping, *Pisciculture* is fish farming, and *Aquaculture* is farming aquatic organisms.

34. **Answer: A)** Proteins

- Explanation: Both silk and wool are animal fibers. Wool is made of the protein keratin, while silk is primarily made of the proteins fibroin and sericin.

35. **Answer: D)** 3000 B.C.

- Explanation: According to historical records and legends, silk production began in China around 2700–3000 B.C. during the reign of the Yellow Emperor.

36. **Answer: B)** Bombay silk

- Explanation: Tassar, Muga, and Eri are all well-known natural varieties of silk produced by different species of silkworms. "Bombay silk" is not an official structural variety of silk fiber.

37. **Answer: D)** Sericulture

- Explanation: As noted in question 33, the dedicated practice and science of breeding and rearing silkworms for commercial silk harvesting is called sericulture.

38. **Answer: A)** Bombyx mori

- Explanation: *Bombyx mori* is the domestic silk moth variety responsible for producing the vast majority of commercial mulberry silk.

39. **Answer: A)** Steel

- Explanation: A thread of natural silk fiber is remarkably strong; its tensile strength is comparable to that of a steel wire of the same thickness.

40. **Answer: A)** Sorting

- Explanation: Sorting is the step where the washed (scoured) wool is carefully separated and categorized based on factors like length, texture, and fiber size.

LEVEL – II

1. Correct option: (A)

Rearing of sheep means to look after the sheep by providing them with food, shelter and healthcare. It involves the management and care of the sheep to ensure their well-being and productivity.

2. Correct option: (B)

The persons who look after the sheep are called shepherds. They look after herding and caring for the sheep, ensuring that they have access to food, water, and shelter, as well as protecting them from predators and guiding them to suitable grazing areas. Shepherds play a crucial role in sheep farming and ensuring the well-being of the flock.

3. Correct option: (A)

Sheep are herbivores, meaning they primarily feed on plant material. They are animals, equipped with a four-chambered stomach that allows them to digest fibrous plant material efficiently. Sheep graze on grass, leaves, and other vegetation.

4. Correct option: (A)

In winter, sheep are often kept indoors or in closed sheltered areas to protect them from the cold weather. Sheep are susceptible to adverse weather conditions, specially extreme cold and wetness, which can cause health issues and even death in severe cases. By keeping them indoors or providing them with shelter, farmers can ensure that the sheep are safe, warm, and protected from the harsh winter elements.

5. Correct option: (A)

Cutting of wool coat of sheep along with thin layer of skin is called as **Fleece**.

6. Correct option: (A)

Shearing is the process of removing the wool from the body of a sheep in the form of a fleece. This is essential for the health and well-being of the sheep as it prevents them from becoming too hot during warm weather and reduces the risk of certain skin conditions. The shearing process is typically performed by skilled workers called shearers or wool harvesters. Shearing allows farmers to collect wool, which is a valuable natural fiber used to produce cloths in various textiles and products.

7. Correct option: (A)

The process of washing the fleece (cut hair of sheep) to remove dust, dirt, dried sweat, and grease is called scouring. Scouring is an essential step in processing of raw wool, as it prepares the wool for further manufacturing into various products such as yarn, fabric, and garments.

These impurities include lanolin which is a natural oil secreted by the sheep's skin, dust and dirt from the environment, and sweat residues. If not properly removed, these impurities can affect the quality and appearance of the final wool product.

8. Correct option: (C)

The fleece of sheep is washed with water and soap during the scouring process. Soap helps to remove dirt, dust, dried sweat, and grease from the wool fibers, leaving the fleece clean and ready for further processing. The use of soap in the scouring process ensures that impurities are effectively removed, and the wool is prepared for manufacturing into various woolen products like yarn, fabric, and garments.

9. Correct option: (A)

The process of separating the fleece of a sheep into sections according to the quality of woolen fibers is called sorting. After shearing and before the scouring process, the raw wool is sorted to classify the fibers based on characteristics such as fiber length, fineness, color, and overall quality.

Sorting is an essential step in the wool processing chain, as it allows the wool to be categorized into different grades, each suitable for specific products.

10. Correct option: (D)

Silk fiber is made of protein. It is primarily composed of a protein called fibroin, which makes up the important part of the silk fiber. The fibroin protein provides silk with its unique properties such as strength, smoothness, and luster.

Silk is produced by various insects, but the most well-known and widely used silk is produced by the silkworm (Bombyx mori). The silkworm spins a cocoon made of silk fibers to protect itself during the pupal stage of its life cycle. The silk fibers are produced from specialized glands in the silkworm's head called spinnerets.

11. Correct option: (B)

Silk is one of the strongest natural fibers. It has an impressive tensile strength, means it can withstand considerable pulling or stretching forces without breaking. Despite its delicate and smooth appearance, silk is surprisingly durable and has been historically used in various applications, including garments, textiles, and even parachutes in some cases.

12. Correct option: (C)

Sericulture is an ancient practice that involves the cultivation of silkworms and the harvesting of their cocoons to produce silk fibers. It is an essential part of the silk production process.

Sericulture involves several stages, starting with the production of silkworm eggs, which are carefully incubated to hatch into larvae. The larvae are then fed with mulberry leaves. They undergo several molting stages, during which they shed their skin and grow larger.

After reaching the final larval stage, the silkworms spin a cocoon made of silk fibers to protect themselves during the pupal stage of their life cycle. The cocoons are collected and then processed to obtain the raw silk fibers.

13. Correct option: (A)

The silky covering spun by the larva of silkworm, is called a cocoon. When the silkworm is ready to undergo the pupal stage of its life cycle, it spins a protective casing around itself using silk fibers produced from specialized glands in its head. This cocoon serves to protect the silkworm during its transformation into a pupa, and later, into an adult moth.

14. Correct option: (D)

The cocoon is made by the silkworm to protect its development as a pupa. After going through several larval stages, the silkworm spins a silk cocoon around itself during its pupal stage. Inside the cocoon, the silkworm undergoes metamorphosis and transforms into an adult moth. The cocoon serves as a protective casing that shields the pupa from external threats until it is ready to emerge as an adult moth.

15. Correct option: (C)

The breaking of the cocoon lowers the quality of silk. When the cocoon is intact and not broken, the silk fibers inside it can be carefully unraveled in long continuous filaments. This results in high-quality silk, which is valued for its strength, sheen, and smooth texture.

However, if the cocoon is broken, the long silk filaments are cut into shorter segments, which reduces the overall quality of the silk. Silk with shorter fibers may

be less durable and have a rougher texture, making it less desirable for certain high-end silk products.

16. Correct option: (B)

The process of taking out silk fibers from the cocoons for use as silk is called reeling. Reeling is an essential step in the silk production process and involves carefully unwinding the silk filaments from the cocoons without breaking them.

During reeling, the cocoon is softened by exposing it to heat or steam. Then, skilled workers gently and carefully unwind the silk filaments from the cocoon, sometimes using a special tool called a reel.

17. Correct option: (A)

The most common silk moth is Mulberry silk moth, scientifically known as Bombyx mori. Mulberry silk is widely produced and is obtained from the domesticated silkworms of the species **Bombyx mori**.

These silkworms feed primarily on the leaves of the mulberry tree, which is why the silk produced by them is called Mulberry silk. The cultivation of silkworms for Mulberry silk production is a practice known as sericulture.

18. Correct option: (D)

Rayon is a man-made fiber that is often used as a substitute for natural silk in textiles and clothing. It is produced from cellulose, usually obtained from wood pulp, and then chemically processed to create long filaments that are spun into threads.

19. Correct option: (B)

The burning test is one of the methods used to distinguish difference between natural silk and artificial silk. This test involves burning a small sample of the fabric to observe its behavior and characteristics during combustion.

Natural Silk (Pure Silk): When natural silk burns, it will have a smell similar to burning hair or feathers. The flame will be steady. Once the flame is removed, the silk will stop burning on its own. The ash left behind will be very fine and powdery, often white or light gray in color.

Artificial Silk (Rayon): When rayon burns, it may produce a chemical or sweet smell. The flame will burn steadily and is easier to maintain compared to natural silk. After the flame is removed, rayon will continue to burn for a short time. The ash left behind is typically darker and may be bead-like.

20. Correct option: (C)

China is the leading manufacturer of silk in the world. The country has a long history of silk production, dating back thousands of years. Chinese silk is highly regarded for its quality. China's favorable climate, increase supply of mulberry leaves (the primary food of silkworms), and expertise in silk production have contributed to its dominant position in the global silk industry.

21. Answer: A,B,C) It is obtained from the fleece of sheep

- Explanation: Wool is primarily harvested from the protective winter coat (fleece) of sheep. While wool is soft and has natural crimps (curls), statement C is the most defining and fundamentally accurate description. Woolen clothes keep us *warm*, not cool.

22. **Answer: D)** All of these

- Explanation: Although sheep fleece is the most common source, the specialized under-fleece of camels (like Alpaca and Llama) and even specific kinds of dog hair (like Chiengora from specific breeds) can be spun into wool-like yarn.

23. **Answer: A,B)** Ladakh (Also common in Tibet)

- Explanation: Yaks thrive in freezing, high-altitude alpine climates. In India, yak wool is traditionally harvested and used in the cold Himalayan regions of Ladakh and Tibet.

24. **Answer: A,B,C,D**

- Explanation: *Shearing* is the specific initial process of shaving the thick fleece off the sheep's body. Scouring, sorting, and dyeing are subsequent chemical or processing stages to convert that harvested fleece into usable yarn.

25. **Answer: A,B)** Alpaca (Note: Alpaca is globally famous for its premium wool fleece). Both A and B are technically South American wool yielders, but Alpaca is primarily bred for high-quality commercial wool.

26. **Answer: A, B, and C**

- Explanation: Sheep, goats (like Angora and Cashmere goats), and yaks all provide fleece used to spin commercial wool. Pigeons are birds with feathers, so they do not produce wool.

27. **Answer: A, B, C, and D** are all Indian breeds/regions! *
Explanation: Lohi (Rajasthan/Punjab), Rampur Bushair (Himachal Pradesh), and Nali (Haryana/Punjab) are prominent registered indigenous Indian sheep breeds.

28. **Answer: A, B, C, and D are all varieties! ***
Explanation: Mulberry silk, Tassar silk, Eri silk, and Muga silk represent the major natural commercial silk varieties produced by different silk moth species.

Assertion and Reason Type

29. **Assertion:** The fur on the body of a camel is also used as wool. Reason: Weaving yarn makes a piece of fabric.

- **Answer: B)** Both A and R are correct, but R is not the correct explanation of A
- Explanation: Both facts are true independently: camel fur is used to make luxurious wool (such as alpaca or camel hair fabric), and weaving transforms yarn into fabric. However, the mechanism of making a fabric does not explain *why* camel fur works as a wool fiber.

30. **Assertion**: Sheep have a thick coat of hair on their body. Reason: This is to keep sheep warm.

- **Answer: A)** Both A and R are correct, and R is the correct explanation of A
- Explanation: The thick fleece traps a significant layer of still air close to the sheep's skin. Since air is a poor conductor of heat, it prevents body heat from escaping, keeping them warm in cold environments.

31. **Assertion**: The process of removing seed from cotton called ginning. Reason: Spinning is a process of making fibres.

- **Answer: C)** A is correct, but R is incorrect
- Explanation: The assertion is true; ginning separates cotton seeds from the raw fibers. The reason is false because spinning is the process of twisting fibers into yarn, not making fibers themselves.

32. **Assertion:** Polyester is a natural fibre. Reason: Muga is the strongest variety of silk.

- **Answer: D)** A is incorrect, but R is correct
- Explanation: Polyester is completely synthetic (man-made), making the assertion false. Muga silk, native to Assam, is globally renowned for its exceptional tensile strength and natural golden luster, making the reason true.

Matrix Matching Type

33. **Answer: a) → S, b) → P, c) → Q, d) → R**

- Breakdown Table:

Column I	Column II	Explanation
a) Scouring	S) Cleaning sheared skin	Scouring washes away dirt, dust, and grease.
b) Cocoon	P) Yields silk fibres	The silk protective shell spun by the silkworm.
c) Yak	Q) Wool yielding animal	Common wool source in cold alpine regions.
d) Mulberry leaves	R) Food of silkworm	The exclusive primary diet of *Bombyx mori*.

Comprehension Type

34. **Answer: B)** Animals

- Explanation: Wool is a natural protein-based fiber harvested directly from living animals.

35. **Answer: C)** Fine soft under hair

- Explanation: While sheep have coarse beard hair, it is the soft, crimped *under-hair* closest to the skin that provides high-quality textile wool.

36. **Answer: B)** Wool

- Explanation: As stated in the provided text snippet, Yaks are traditional cold-climate fleece-bearing animals that provide wool.

Previous Year Questions (NSO)

1. **Answer: A. Silk**

- Explanation: Silk is celebrated as one of the strongest natural proteins, possesses a brilliant natural shine, and commercial harvesting involves boiling cocoons containing the pupae inside.

2. **Answer: D**. Both B and C

- Explanation: Reeling is the industrial extraction of delicate silk filaments from cocoons (B is true), and Mulberry leaves are eaten by silkworms (C is true). *Shearing* is cutting the hair, not washing it with soap (which is called scouring).

3. **Answer: C**. Lohi – Uttar Pradesh

- Explanation: The Lohi breed of sheep is indigenous to Rajasthan and Punjab, not Uttar Pradesh. The other pairs are correctly matched to their native regions.

4. **Correct Option Combination: C. F, F, T**

5. **Answer: A.** Both statements 1 and 2 are true and statement 2 is the correct explanation of statement 1.

- Explanation: Air is a poor conductor of heat. The structural air pockets within wool fiber trap a layer of dead air, preventing metabolic body heat from radiating away into cold environments.

6. **Correct Option Combination:** All statements are True. *(Note: If this option alignment matches your exam sheet keys, verify if your choices contain a T, T, T alternative).*